

Ultraslow serotonin oscillations in the hippocampus delineate substates across NREM and waking

Reviewed Preprint

v2 • June 19, 2025

Revised by authors

Reviewed Preprint

v1 • September 20, 2024

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This **important** study demonstrates that slow fluctuations in serotonin release during wakefulness and non-REM sleep correspond to periods of heightened arousal or enhanced offline information processing. The evidence supporting this claim is **convincing**, and the methodology is robust and broadly applicable, likely to benefit many researchers in the field. This work will be of significant interest to neuroscientists studying sleep, memory, and neuromodulation.

<https://doi.org/10.7554/eLife.101105.2.sa4>

Abstract

Beyond the vast array of functional roles attributed to serotonin (5-HT) in the brain, changes in 5-HT levels have been shown to accompany changes in behavioral states, including WAKE, NREM and REM sleep. Whether 5-HT dynamics at shorter time scales can be seen to delineate substates within these larger brain states remains an open question. Here, we performed simultaneous recordings of extracellular 5-HT using a recently-developed G Protein–Coupled Receptor–Activation–Based 5-HT sensor (GRAB5-HT3.0) and local field potential (LFP) in the hippocampal CA1, which revealed the presence of prominent ultraslow (<0.05 Hz.) 5-HT oscillations both during NREM and WAKE states. Interestingly, the phase of these ultraslow 5-HT oscillations was found to distinguish substates both within and across larger behavioral states. Hippocampal ripples occurred preferentially on the falling phase of ultraslow 5-HT oscillations during both NREM and WAKE, with higher power ripples concentrating near the peak specifically during NREM. By contrast, hippocampal-cortical coherence was strongest and microarousals and intracranial EMG peaks were most prevalent during the rising phase in both wake and NREM. Overall, ultraslow 5-HT oscillations delineate substates within the larger behavioral states of NREM and WAKE, thus potentially temporally segregating internal memory consolidation processes from arousal-related functions.

Introduction

The impact of the outside world on neural activity is highly dynamic and dependent on the state of the brain. During waking behavior, sensory stimuli are actively processed by the brain and shape ongoing brain activity, whereas during sleep, the impact of such external stimuli is reduced in favor of internally-generated rhythms. Transition among behavioral states is accompanied by changes in the extracellular levels of neuromodulators. One such neuromodulator, serotonin (5-HT), shows clear state-dependent changes in activity, with the firing of 5-HT neurons in the brainstem being highest during waking, intermediate during NREM, and lowest during REM states (McGinty and Harper 1976 [↗](#), Trulson and Jacobs 1979 [↗](#)). Furthermore, changes in 5-HT levels have been causally linked to brain state changes, though some controversy over the direction of such changes remains. While some studies suggest a wake-promoting role for 5-HT, others propose that 5-HT increases sleep drive over the course of waking (Ursin 2008 [↗](#), Monti et al. 2008 [↗](#)). In either case, state-dependent changes in 5-HT levels can be seen to reorganize brain networks in response to ongoing functional demands.

Beyond traditional brain states, recent attention has been drawn to the existence of substates within these larger brain states. NREM sleep, for example, has been shown to contain periods of high and low arousal (Lecci et al. 2017 [↗](#), Osorio-Ferero et al. 2021 [↗](#)). In these studies, high arousal substates were associated with higher heart rate and sensitivity to auditory stimuli, while low arousal substates contained more hippocampal ripples and sleep spindles. Therefore, substates in NREM may mediate the balance between processing external stimuli and carrying out internal brain processes, such as memory consolidation. Importantly, in both studies, these substates were delineated by the phase of ultraslow oscillations (< 0.1 Hz.) of sigma power and noradrenaline levels, respectively. While NREM substates have not yet been examined in relation to 5-HT levels, ultraslow oscillations have been observed in population activity in the dorsal raphe nucleus (DRN) (Kato et al. 2022 [↗](#), Milner et al. 2016), as well as in extracellular 5-HT levels in the hippocampal dentate gyrus (Turi et al. 2024 [↗](#)) during NREM, suggesting that 5-HT may also distinguish pro-arousal and pro-memory substates.

5-HT is a key modulator for many brain functions, which is reflected by the highly extensive projections of serotonergic fibers throughout the mammalian brain. Especially dense are the connections from the midbrain raphe nuclei, the source of 5-HT to the hippocampus, a region important for memory processing (Jacobs and Azmitia 1992 [↗](#)). Hallmarks of the hippocampus, ripples are transient fast oscillations (120-250 Hz) observed in the local field potential (LFP) and have been shown to functionally underlie memory consolidation and replay (Buzsàki 2015). While the contribution of 5-HT to memory processing remains unclear, with different studies supporting facilitating vs. suppressing roles (Meneses 2015 [↗](#), Corey and Quednow 2022, Teixeira et al. 2018 [↗](#), Zhang et al. 2013 [↗](#), Zhang et al. 2015, Zhang et al. 2016 [↗](#)), the three studies examining the effect of 5-HT modulation on ripples all found a suppressive effect (Wang et al. 2015 [↗](#), ul Haq et al. 2016 [↗](#), Shiozaki et al. 2023 [↗](#)). However, interpreting these studies is made difficult by the methods used to manipulate 5-HT levels, namely optogenetic activation of 5-HT neurons and pharmacologic interventions at high doses. Simultaneous brain-wide increases of 5-HT levels resulting from optogenetic raphe stimulation and pharmacological interventions via systemic injections, have the potential to activate 5-HT sub-systems which are not naturally active together, such as the reward-activated and movement initiation-activated serotonergic fibers described in the dorsal hippocampus (Luchetti et al. 2020 [↗](#)). Further rendering these studies hard to interpret is the question of physiologically plausible dose, as biphasic dose-dependent effects of 5-HT have been described (Calabrese 2001 [↗](#)).

To bypass such constraints, in the present study we utilized the recently-developed G Protein–Coupled Receptor–Activation–Based (GRAB) 5-HT sensor which allows for the measurement of physiological changes in local extracellular 5-HT concentrations with high spatial and temporal resolution (Deng et al. 2024 [↗](#)). Alongside 5-HT levels, we recorded hippocampal activity with silicon probes in order to examine potential correlations between local 5-HT fluctuations and behavioral states and substates in the dorsal CA1 in freely-moving mice. After simultaneous fiber photometry and LFP recordings, we could identify substates of NREM and WAKE delineated by different phases of ultraslow 5-HT oscillations. These substates roughly corresponded to periods of higher and lower arousal, with lower arousal associated with the preferential occurrence of ripples, and higher arousal with the occurrence of microarousals (MAs) as well as peaks in the intracranial EMG (icEMG, see *Methods: State scoring*) and hippocampal-cortical coherence.

Results

In order to simultaneously record local 5-HT levels and electrophysiological signals, we first injected mice with a virus encoding the GRAB5-HT3.0 sensor (AAV9-hSyn-5HT3.0) in the right dorsal CA1 (Deng et al. 2024 [↗](#)). Simultaneous fiber photometry and electrophysiology recordings were achieved by implanting an optic fiber (400 μ m diameter) above the injection site (**Figure 1A** [↗](#)) and a silicon probe in the left dorsal CA1, at the same anterior-posterior coordinates as the site of viral injection (see *Methods*).

To verify that the sensor reports changes in local 5-HT levels, we treated a subset of mice ($n=3$) with fluoxetine (10 mg/kg), an SSRI known to acutely increase extracellular 5-HT levels in the dorsal hippocampus (Imoto et al. 2015 [↗](#)). Compared to saline, fluoxetine significantly elevated fluorescence recorded by fiber photometry, confirming the sensitivity of the GRAB-5HT3.0 sensor to endogenous levels of 5-HT in the hippocampus of mice (**Figure 1I** [↗](#)).

In order to examine the relationship between 5-HT signals and hippocampal activity across brain states, we simultaneously recorded hippocampal GRAB5-HT3.0 fluorescence and local field potentials (LFP) in the home cage during normal behavior, which included both waking and sleeping bouts. Twelve recording sessions were conducted using six mice (1–3 sessions per mouse), all of which were included in subsequent analysis and statistical testing. Automated sleep-scoring (Watson et al. 2016 [↗](#)) was performed on the LFP data, showing the occurrence of different behavioral states, namely WAKE, non-REM sleep (NREM), REM sleep, and microarousals (MA), with different frequencies throughout the recording sessions (**Figure 1G** [↗](#)). Consistent with prior reports in the hippocampus (Park et al. 1999 [↗](#)), 5-HT levels were highest during WAKE, intermediate during NREM, and lowest during REM (**Figure 1F** [↗](#)). During MAs, 5-HT levels were on average in between WAKE and NREM levels, but statistically indistinguishable from WAKE. Most strikingly, we observed prominent ultraslow (<0.06 Hz.) oscillations of 5-HT levels during NREM (**Figure 1D–E** [↗](#)). These ultraslow oscillations were reflected as a clear peak in the power spectrum at ~ 0.015 Hz., with significantly greater power in the < 0.04 Hz. range than in the > 0.04 Hz. range for both states (**Figure 1H** [↗](#)). Similar ultraslow oscillations were observed during WAKE, though with significantly less power than NREM ultraslow oscillations (NREM / WAKE = 1.21 ± 0.028).

The waveform of these ultraslow 5-HT oscillations was shown to be asymmetric with respect to both slope (**Figure S2A–B** [↗](#)) and length (**Figure S2C–D** [↗](#)), with a steeper and shorter rising phase and a flatter and longer falling phase. Although the GRAB5HT3.0 sensor we use exhibits a similar kinetic asymmetry, with a shorter rise time (0.25 s) than decay time (1.39 s) (Deng et al. 2024 [↗](#)), we consider it unlikely to account for the asymmetry observed in ultraslow 5-HT oscillations. For one, the 5-HT signal clearly contains reductions in 5-HT levels that are much faster than the descending phase of the ultraslow oscillation. Furthermore, the finding that a subset of dorsal raphe neurons firing at ultraslow frequencies display clear asymmetry in rising time versus decay

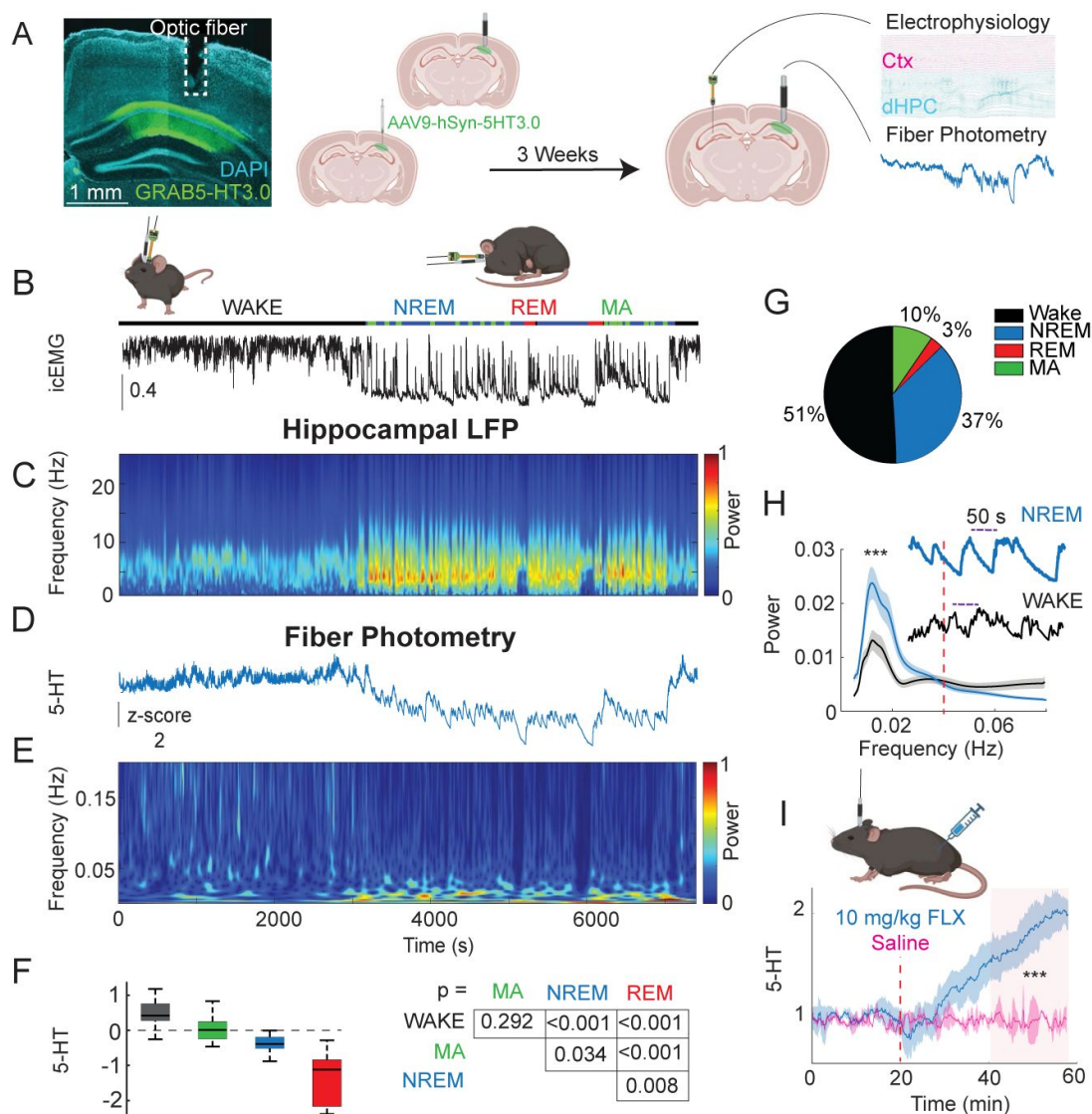


Figure 1.

5-HT levels exhibit ultraslow oscillations during NREM and WAKE.

A. Histology and experimental protocol. Left: expression of GRAB5-HT3.0 sensor (in green) in dorsal CA1 with optic fiber track above. Right: methodology for dual implantation surgeries. AAV9-hSyn-5HT3.0 was first injected into the right dorsal CA1. In the same surgery, an optic fiber was implanted above the injection site. After three weeks of viral expression, a silicon probe was implanted above the left dorsal CA1. Simultaneous recording of the GRAB5-HT3.0 sensor activity (fiber photometry) and electrophysiology was performed. **B-E.** Example dual fiber photometry-electrophysiology recording with times shown in **E**. **B.** Labeled sleep states resulting from automated sleep-scoring above an intracranial EMG trace. **C.** Spectrogram (Stockwell transform) showing normalized power of a hippocampal LFP channel during awake and sleep states. **D.** Z-scored 5-HT trace. **E.** Spectrogram (Stockwell transform) of the 5-HT trace shown in **D**. **F.** Left: Mean 5-HT level by state, across all experiments (total n=6 mice, 12 recording sessions of 1.5-3 hours). Right: p-values from a multiple comparisons test applied after fitting a Bayesian GLMM to the data. **G.** Pie chart showing proportion of time spent in different behavioral states, averaged across all experiments. **H.** Top: examples of ultraslow 5-HT oscillations in NREM and WAKE. Bottom: Power spectrum of 5-HT signals in WAKE vs. NREM sleep. Frequencies were separated into two groups (> 0.04 Hz. and <0.04 Hz.), and a GLMM was fitted to the data. P-values shown were derived from a post-hoc multiple comparisons test on the fitted model. Results indicated higher power in the lower half of frequencies (<0.04 Hz.) for both states ($p < 0.0001$), as well as a significant difference in the lower frequencies between NREM and WAKE ($p < 0.0001$), with a NREM/WAKE ratio of 1.21 ± 0.028 . **I.** Control fluoxetine and saline injection experiments. A significant difference between the post-injection period of saline-injected and fluoxetine-injected animals (shaded in red) was observed (Wilcoxon ranked-sum test, $p < 0.001$, $n = 3$ mice).

suggests that the asymmetry we observe in our data could be due to neural activity rather than temporal smoothing by the sensor (Mlinar et al. 2016 [↗](#)). In this same direction, another study found similar asymmetry in extracellular 5-HT levels measured with fast scan cyclic voltammetry (FSCV), a technique with greater temporal resolution than our sensor, after single pulse stimulation of serotonin fibers (Bunin and Wightman 1998).

Next, we sought to investigate whether substates could be defined relative to the phase of these ultraslow 5-HT oscillations, as previously described for sigma power and noradrenaline oscillations (Lecci et al. 2017 [↗](#), Osorio-Ferero et al. 2021 [↗](#)). The stronger ultraslow oscillations of 5-HT during NREM as compared to WAKE states led us to hypothesize a stronger coupling between ultraslow 5-HT oscillations and hippocampal activity during NREM than during WAKE.

5-HT and ripples

First, we looked at the relationship between ultraslow 5-HT oscillations and hippocampal ripples, the electrophysiological signature of memory consolidation. Given the noted shortcomings of the standard spectral filter-based methods for detecting ripples (Liu et al. 2022 [↗](#)), and the recent surge of papers proposing alternative detection algorithms (Watanabe et al. 2021 [↗](#), Hagen et al. 2021 [↗](#), Navas-Olive et al. 2022 [↗](#), Sebastian et al. 2023 [↗](#)), we chose to detect ripples with a custom convolutional neural network (CNN) model (see *Methods*). 400 ms segments of eight LFP channels, including four neocortical and four hippocampal channels, served as input to the model. After training and validation, the best-performing model consisted of four 2D convolutional blocks followed by two dense layers, outputting a 400-ms ripple probability vector (**Figure 2A** [↗](#)). This output vector was thresholded to detect ripples. Notably, the model was able to successfully distinguish ripples from non-ripple uniformly-propagating fast oscillations and movement-related noise (**Figure 2B, C** [↗](#)). Importantly, the features of detected events fell within the bounds expected for hippocampal ripples, with ripple duration and z-scored power showing a log-normal distribution (Levenstein et al. 2019 [↗](#)) (**Figure 2D** [↗](#)).

In comparing ripple occurrence to the 5-HT signal, we noticed that both peaks of power in the ripple band (120-250 Hz.) (**Figure 2E** [↗](#)), a measure independent from ripple detection, and detected ripples (**Figure 2F** [↗](#)) tended to occur on the falling phase of slow 5-HT oscillations. In order to get a better sense of the timing between the 5-HT signal and ripple occurrence, we extracted ripple clusters, which were defined as groups of 10 or more ripples with an inter-ripple interval of three seconds or less (**Figure 2G** [↗](#)). These parameters were empirically observed to capture the clusters of ripples of various lengths occurring during the falling phase of 5-HT oscillations and to exclude the less numerous, non-clustered ripples occurring on the rising phase. Having defined ripple clusters, we were then able to isolate the first and last ripples of each cluster. When considering all ripples, before cluster extraction, the average ripple was seen to occur on the falling phase of the 5-HT oscillation, in both NREM and WAKE (**Figure 2H1-2, first column** [↗](#)). The first ripple of ripple clusters consistently occurred shortly after the peak in 5-HT in NREM, and at the peak in WAKE, while the last ripple occurred at the trough in both NREM and WAKE (**Figure 2H1-2, columns 2-3** [↗](#)). In summary, ripples were seen to span the length of the falling phase of ultraslow 5-HT oscillations in both NREM and WAKE, though this relationship was generally stronger in NREM.

To further probe the relationship between ripple occurrence and 5-HT, we examined when ripples preferentially occur in relation to the phase of ultraslow 5-HT oscillations. We first looked at the distribution of inter-ripple intervals (IRIs) along different phases of ultraslow 5-HT oscillations measured in the 0.01-0.06 Hz frequency band (**Figure 3A-D** [↗](#)), which showed the strongest temporal coupling to ripples compared to other slow oscillation frequencies (**Figure S3** [↗](#)). There, consistent with what we observed in the cluster extraction data, we found lower IRIs along the falling phase (-180° : 0°), and higher IRIs along the rising phase (0° : 180°) (**Figure 3B,D** [↗](#)). These

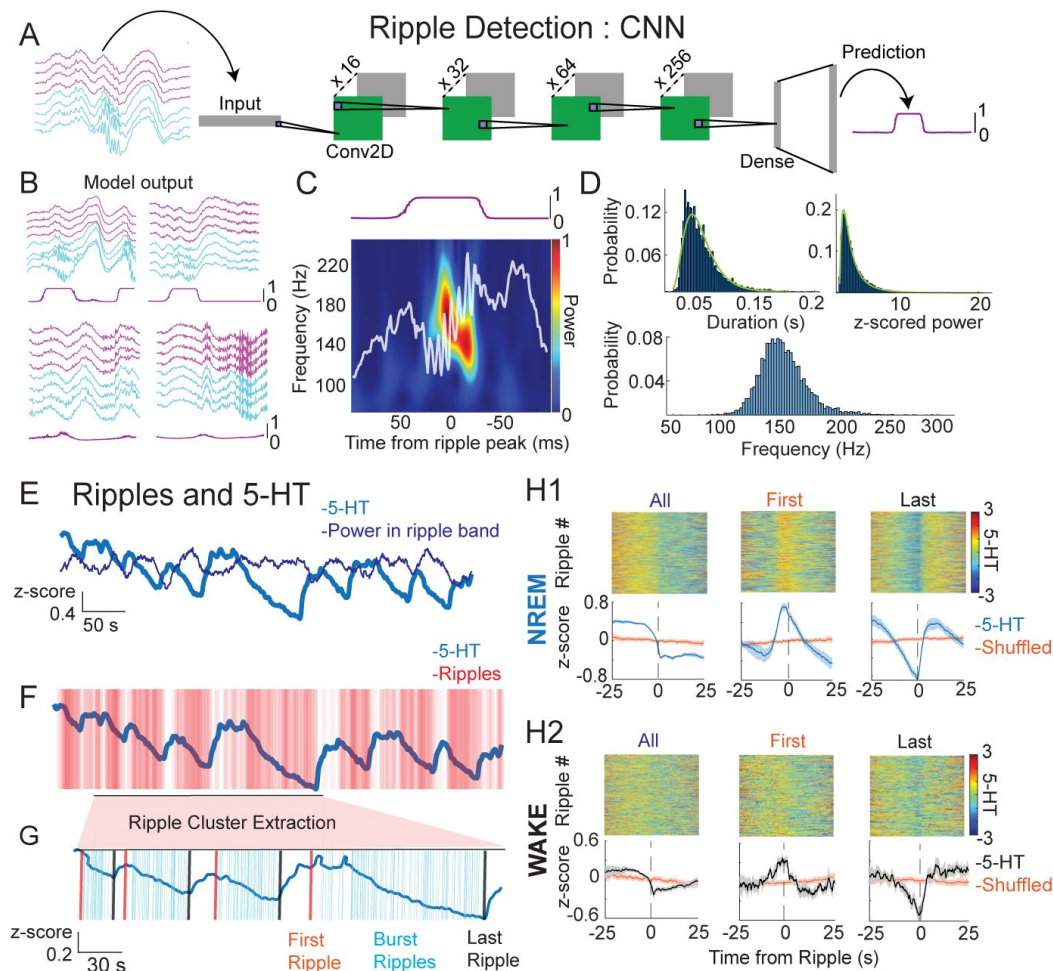


Figure 2.

Ripples occur time-locked to ultraslow 5-HT oscillations.

A. Schematic showing the convolutional neural network used for ripple detection. 8-channel x 400 ms-LFP chunks were used as input. The bottom four channels (cyan) were taken from the dorsal CA1 and contained ripples, and the top four channels (magenta) were chosen from a non-adjacent part of the neocortex above the dorsal CA1. The model consisted of four convolution blocks ('Conv2d'), each block comprising a 2D convolutional layer, a ReLU activation function, and batch normalization. Two dense layers with dropout and batch normalization ('Dense') followed and produced the final output, a 400 ms vector with values between 0-1, indicating the probability of a ripple occurring during the course of the input chunk.

B. Example model output given the four LFP chunk inputs shown. First row: true positives. Second row: fast oscillations and movement artifacts not detected as ripples by the model.

C. Spectrogram from a ripple detected by the model, 0-1 normalized.

D. Characteristics of detected ripples. Ripples from all experiments were included, and probability distributions are shown. Top left: distribution of duration. Top right: distribution of z-score normalized ripple power. Bottom: distribution of ripple frequency. Ripple duration and normalized ripple power follow a log-normal distribution (duration: χ^2 (df = 7, $N = 49,458$) = 1.398e+03, $p < .0001$, normalized ripple power: χ^2 (df = 7, $N = 49,458$) = 422.1862, $p < .0001$).

E. Example 5-HT trace and computed power in the ripple band (120-250 Hz).

F. Same example 5-HT trace and individual detected ripples.

G. Example of ripple cluster extraction. Ripple clusters were defined as having a minimum of 10 ripple events and an inter-ripple interval of less than 3 seconds. Note the few ripples occurring during the rising phase of 5-HT ultraslow oscillations in **F** are excluded from extracted ripple clusters in **G**. From these ripple clusters, the first (orange) and last (black) ripples in a cluster were extracted.

H1-2. Ripple-triggered 5-HT in NREM (**H1**) and WAKE (**H2**) states. The first rows of **H1** and **H2** show all 50 s 5-HT segments centered around the ripple peak for different combinations of ripples (columns). In the first column, all ripples in the given state were included. The second and third columns used only the first or last ripple in extracted ripple clusters, respectively. The second rows of **H1** and **H2** show the mean ripple-triggered 5-HT traces (blue) and randomly shifted traces (orange) for each group of ripples. The orange traces were obtained by randomly shifting the ripple times for each condition and averaging the resulting 5-HT 50 s segments centered around those shifted times.

differences in IRIs between rising and falling phases were statistically significant for both NREM and WAKE (**Figure 3C**). As observed previously, the phase preference of NREM ripples was stronger than that of WAKE ripples ($p < 0.0001$), though both were prominent.

Next, we calculated the ultraslow 5-HT phases at which individual ripples occurred during both NREM and WAKE (3E-F), where we again observed a significant falling phase preference in both states, albeit again stronger in NREM than WAKE ($p < 0.0001$), confirmed by a GLMM where phase preference was modeled as a binary outcome (rising vs. falling) (**Figure 3F2, 3F4**). Despite more session-level variance in the preferred phase angles of WAKE ripples (**Figure S3**), the overall mean phase angles of NREM and WAKE ripples were very similar, at 101.1° and 99.6° , respectively, reflecting a very similar distribution of ripples along 5-HT ultraslow oscillations across states (**Figure 3E3**). This result appears to be at odds with the mean ripple timing relative to 5-HT in NREM shown in **Figure 2H1, column 1**, where the mean ripple falls closer to the 5-HT trough. However, this apparent discrepancy can be explained by the method used to determine the mean ripple timing. In **Figure 2H**, the analysis is agnostic to the frequency of the underlying serotonin oscillation, meaning data is being averaged in segments with different oscillation periods. This leads to some imprecision in the estimation of the exact location along the falling phase where the mean ripple lies. In **Figure 2H**, columns 2 and 3, this is less of a problem, because the serotonin segments are ‘anchored’ by the peak and trough, respectively, as the onsets and offsets of ripple clusters have a clear preference for these time points. Therefore, the fact that the length of the falling phase varies is less important. In **Figure 3F2-4**, the preferred phase for each ripple is calculated on data filtered in the ultraslow frequency band (0.01–0.06 Hz.), so noise from faster or slower rhythms is not included. Phase estimation then accounts for the variability in oscillation period, which within ultraslow frequencies can range from 16.7 to 100 seconds. In summary, the phase analysis likely provides a more precise estimate of the preferred location of ripples within serotonin ultraslow oscillations, however, given that the distribution does not contain a significant peak, but is closer to a uniform distribution within the falling phase, the mean ripple is less meaningful than the general distribution of ripples along the rising and falling phase.

Finally, we wondered whether ripples with different features were preferentially distributed according to the phase of ultraslow 5-HT oscillations. To this end, we first calculated the peak power for each detected ripple. Unlike what we previously observed with respect to ripple occurrence, the power of ripples in NREM showed a significant preference for the peak of ultraslow 5-HT oscillations, whereas no preference was observed in WAKE (**Figure 3F-G**). In summary, while ripples tend to show a preference for the falling phase of ultraslow 5-HT oscillations, stronger ripples tend to be statistically more likely near the peak of 5-HT in NREM, but not WAKE. We then performed the same analysis, looking this time at ripple duration and frequency with respect to ultraslow 5-HT phase. While ripple frequency was highest during the falling phase of ultraslow 5-HT oscillations (**Figure S4A-B**), ripple duration showed no clear phase preference (**Figure S4C-D**).

5-HT and microarousals

Given that ripples, the electrophysiological signature of memory consolidation, were shown to constitute one substate occurring during the falling phase of ultraslow 5-HT oscillations, we next looked for signs of arousal-associated substates, potentially occurring at different phases of the ultraslow 5-HT oscillation. To this end, we first considered the occurrence of MAs relative to 5-HT, given that MAs themselves constitute periods of heightened arousal within NREM (Halász 1998, Watson et al. 2016). MAs were observed together with peaks in the icEMG during NREM, which appeared to be time-locked to 5-HT ultraslow oscillations (**Figure 4A**). On average, MAs occurred shortly before the peak of 5-HT on the rising phase (**Figure 4D1-2**). The same trend was observed when looking at the ultraslow 5-HT oscillation phase at which individual MAs occurred, where a significant preference for the falling phase was identified (**Figure 4C**), with the added information that MAs were generally much more likely on the rising phase than the falling phase.

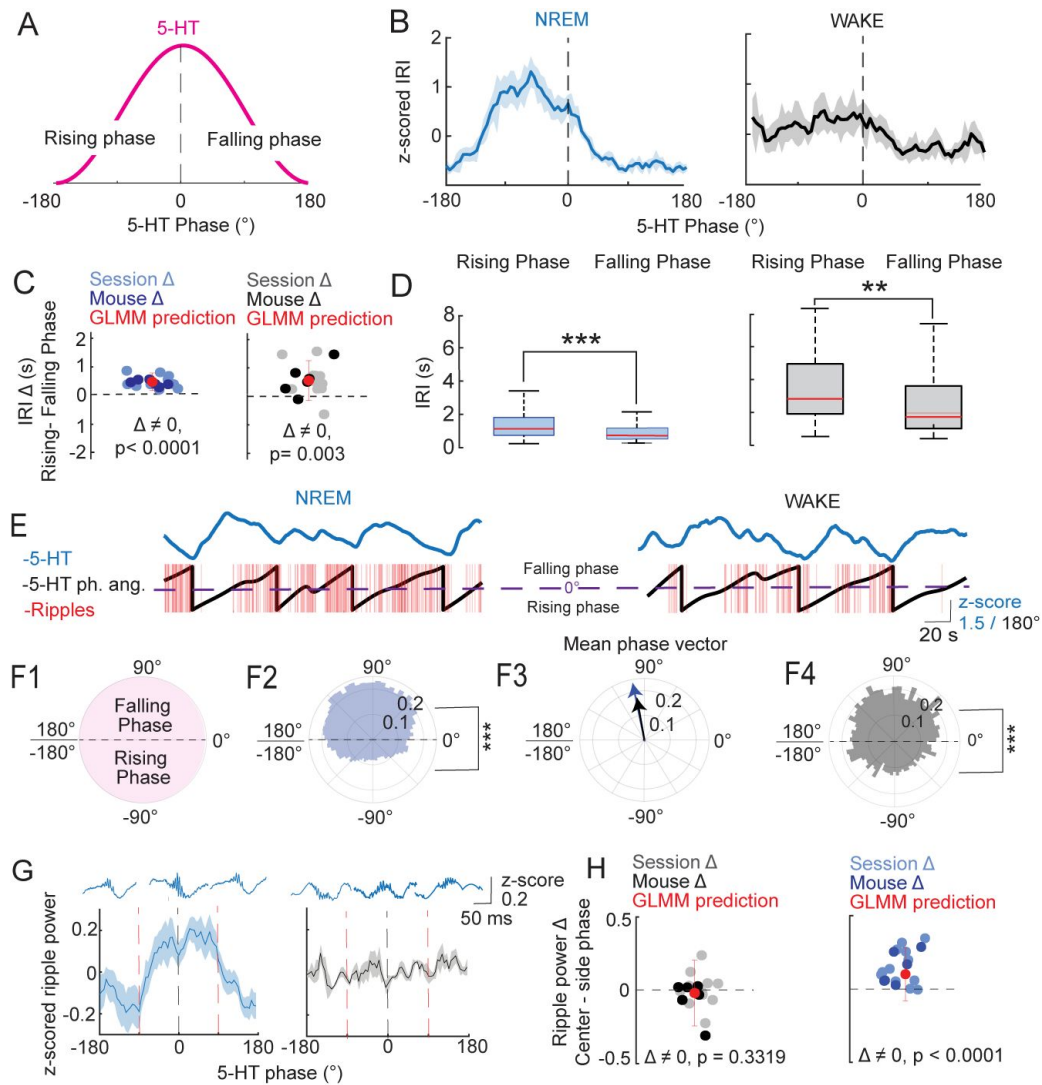


Figure 3.

Ripple occurrence and power vary by the phase of ultraslow 5-HT oscillations.

A. Schematic showing one period of a slow 5-HT oscillation. The rising phase of the oscillation occurs from -180° to 0° , and the falling phase occurs from 0° to 180° . B. Mean z-scored inter-ripple interval (IRI) by 5-HT phase angle during NREM (left) and WAKE (right). C. Mean rising phase IRI - mean falling phase IRI, plotted by session and mouse level in WAKE (left) and NREM (right). Red point with error bar indicates predicted difference and confidence interval after fitting a GLMM to the data. P-values shown were derived from a post-hoc multiple comparisons test on the fitted model. (n=6 mice, 12 sessions). D. Box plots showing IRI in rising vs. falling phase in NREM (left) and WAKE (right). Significance levels shown reflect results of statistical analysis in C. E. Example 5-HT trace (top) and corresponding 5-HT phase angles and ripples (bottom) for NREM (left) and WAKE (right). The peak of the ultraslow oscillation (0°) is indicated by the dashed purple line. F1. Schematic polar plot showing one period for a slow 5-HT oscillation. The falling phase of the oscillation occurs from 0° to 180° , and the rising phase occurs from -180° to 0° . F2. Phase of all NREM ripples relative to the ultraslow 5-HT oscillation. A significant ($p < 0.0001$) bias toward the falling phase was found after fitting a GLMM to the data with a binomial link function. F3. Mean phase vector of NREM and WAKE ripples. A significant ($p < 0.0001$) bias toward the falling phase was found after fitting a GLMM to the data with a binomial link function. F4. Phase of all WAKE ripples relative to the ultraslow 5-HT oscillation. G. Z-scored ripple power by 5-HT phase angle during NREM (left) and WAKE (right). Red vertical dashed lines delineate analyzed phase segments: 'center' (-90° to 90°) vs. 'side' (-180° to -90° and 90° to 180°). Representative ripples from each phase grouping are shown above. H. Mean center phase ripple power - mean side phase ripple power, plotted by session and mouse level in WAKE (left) and NREM (right). Red point with error bar indicates predicted difference and confidence interval after fitting a GLMM to the data. P-values shown were derived from a post-hoc multiple comparisons test on the fitted model.

Therefore, not only do MAs themselves define periods of arousal within NREM, but their occurrence is biased by the phase of the ultraslow 5-HT oscillation, which designates periods when such arousals can occur.

5-HT and icEMG

As MAs are only observed during NREM, we could not perform the same analysis on WAKE data. However, given that MAs are accompanied by peaks in the icEMG, we next examined whether the icEMG itself is time-locked to the phase of ultraslow 5-HT oscillations across states. Not surprisingly given the MA results, icEMG peaks during NREM occurred preferentially during the rising phase of ultraslow 5-HT oscillations (**Figure 4E,F**, left). Interestingly, icEMG peaks during WAKE were also observed preferentially in the rising phase (**Figure 4E,F**, right). The difference in icEMG between the rising and falling phase was shown to be significant in both states, with the effect in NREM being stronger than that in WAKE (**Figure 4G**). Therefore, the rising phase of ultraslow 5-HT oscillations can be seen to constitute a substate of heightened arousal, both in terms of MA occurrence during NREM, and the icEMG itself during both NREM and WAKE states.

5-HT and hippocampal-cortical coherence

Coherence is a measure of synchrony between two brain areas thought to underlie neural communication (Fries 2005). Changes in coherence, including hippocampal-cortical coherence, have been found to track changes in arousal, both across and within brain states (Bastuji et al. 2021, Klaver et al. 2023, Cantero et al. 2004). We therefore examined hippocampal-cortical coherence in relation to the phase of ongoing ultraslow 5-HT oscillations (**Figure 5A**). After computing coherence between pairs of hippocampal and cortical channels in both NREM and WAKE, we observed that in certain frequency bands, coherence was higher in the rising phase of ultraslow 5-HT oscillations than in the falling phase, in both states (**Figure 5B**). In order to more closely examine this trend, we looked at coherence by ultraslow 5-HT oscillation phase for each frequency band individually (**Figure 5C**). We found significant differences in coherence by ultraslow 5-HT phase in theta, slow gamma, fast gamma and high frequency oscillation bands, but not in the delta band both in NREM and WAKE (**Figure 5D**). Thus, inter-areal neural communication seems to be gated by the phase of ultraslow 5-HT oscillations, whereby during the rising phase, coherence is higher and such communication is favored.

Given these results, one could posit that the higher hippocampal-intracortical coherence we observe in the rising phase is due to a lack of activity during this time period, as no activity is easier to ‘synchronize’ than bouts of high activity. In fact, this potential relative silence in the rising phase would be in line with the observation that increases in DRN population activity are associated with decreased wideband power during sleep (Kato et al. 2022). To address this question, we looked into both the power of different hippocampal oscillations and unit activity relative to ultraslow 5-HT oscillations. While increased power during the falling phase was found for all frequency bands during NREM, during WAKE gamma band power deviated from this pattern (**Figure S5A-C**). Specifically, low gamma showed no phase preference, whereas high gamma power was highest around the peak of ultraslow 5-HT oscillations. As hippocampal-cortical coherence was higher during the rising phase in both NREM and WAKE, it is not likely that the relative lack of activity in the rising phase, which is not found broadly in WAKE, is behind it. Furthermore, units with a specific phase preference for the rising phase of ultraslow 5-HT oscillations represented a significant, if less numerous, group than units with a falling phase preference (**Figure S6**).

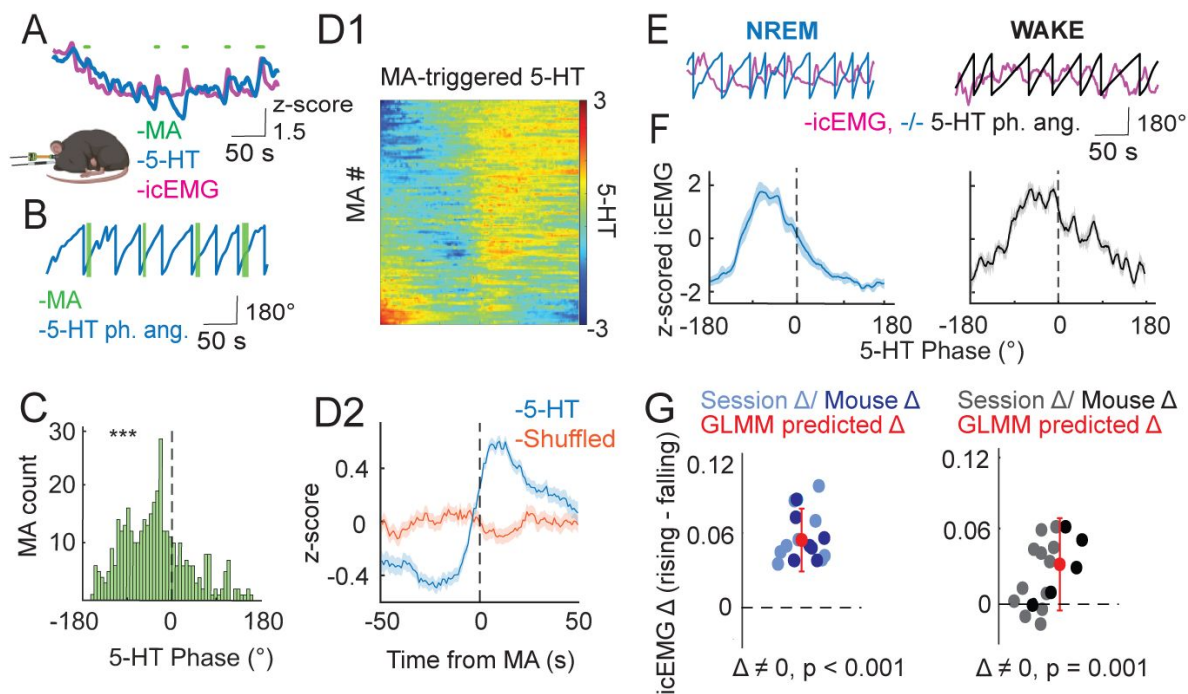


Figure 4.

EMG and MAs vary by the phase of ultraslow 5-HT oscillations.

A.-D2. Relationship between microarousal (MA) occurrence and the phase of slow 5-HT oscillations. **A.** Example trace showing 5-HT, EMG, and MAs during a NREM bout. **B.** Example trace showing extracted 5-HT phase angle and MAs. **C.** MA occurrence according to 5-HT phase angle. A significant ($p < 0.0001$) bias toward the falling phase was found after fitting a GLMM to the data with a binomial link function. **D1.** MA-triggered 5-HT across all MA events. **D2.** Mean MA-triggered 5-HT trace (blue) plotted with mean of randomly shifted 5-HT trace (orange). The orange trace was derived by randomly shifting all MA times and averaging the resulting 5-HT segments around those shifted times. **E.-G.** Relationship between the EMG signal and phase of slow 5-HT oscillations. **E.** Example traces showing extracted 5-HT phase angle and the EMG signal during NREM (left, blue) and WAKE (right, black) states. **F.** Mean z-scored EMG signal by 5-HT phase angle during NREM and WAKE states. **G.** Mean rising phase EMG - mean falling phase EMG, plotted by session and mouse level. Red point with error bar indicates predicted difference and confidence interval after fitting a GLMM to the data. P-values shown were derived from a post-hoc multiple comparisons test on the fitted model.

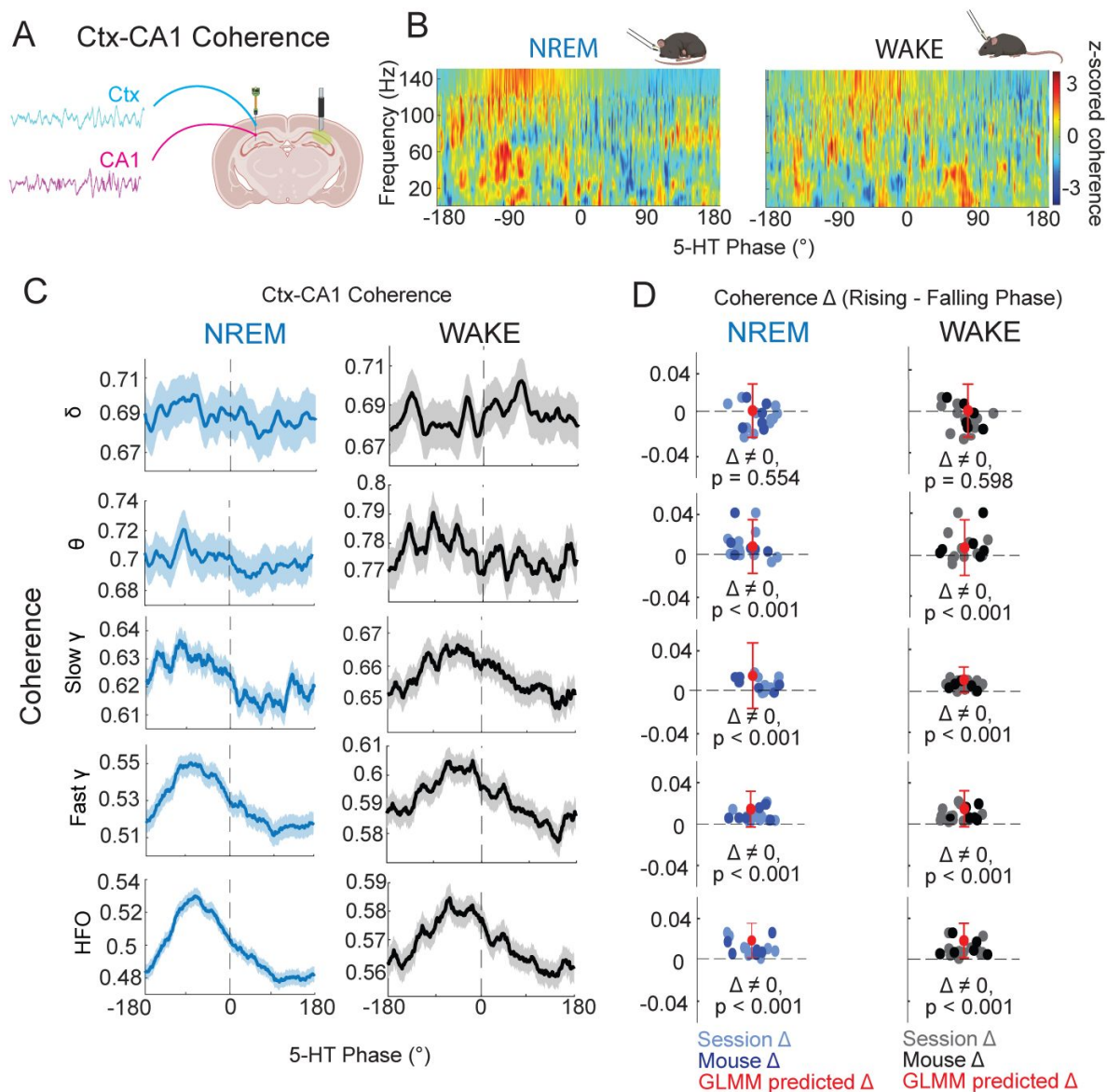


Figure 5.

Coherence varies by the phase of ultraslow 5-HT oscillations.

A. Schematic showing representative hippocampal and cortical traces used for coherence calculations. **B.** Mean z-scored hippocampal-cortical coherence by frequency for NREM (left) and WAKE (right). **C.** Mean coherence by 5-HT phase angle for delta (1-5 Hz.), theta (6-10 Hz.), slow gamma (30-60 Hz.), fast gamma (60-100Hz.) and high frequency oscillation (HFO, 100-150 Hz.) bands in NREM (left column) and WAKE (right column). **D.** Mean rising phase coherence - mean falling phase coherence, plotted by session and mouse level for different frequency bands (rows) and states (columns). Red point with error bar indicates predicted difference and confidence interval after fitting a GLMM to the data. P-values shown were derived from a post-hoc multiple comparisons test on the fitted model.

Discussion

After simultaneous recordings of hippocampal 5-HT levels and LFP across behavioral states, we observed prominent ultraslow oscillations of 5-HT, which timed the occurrence of several electrophysiological read-outs. Specifically, we found that substates, similar to those described in NREM by previous studies (Lecci et al. 2017 [↗](#), Osorio-Ferero et al. 2021 [↗](#)), were closely linked to the phase of 5-HT ultraslow oscillations. Hippocampal ripples occurred preferentially during the falling phase of 5-HT oscillations, whereby hippocampal-cortical coherence was strongest and microarousals and icEMG peaks were most prevalent during the rising phase. Importantly, these 5-HT-defined substates were observed to coordinate local and global brain activity not only within NREM, but also during WAKE states.

The prominence of internally-driven ultraslow 5-HT oscillations during NREM explains why many studies focus on these rhythms during sleep. Potentially due to the requirement of WAKE to integrate external signals, ultraslow 5-HT oscillations appear less prominently in waking behavior. Nevertheless, our data shows that these ultraslow rhythms also play a role in WAKE, albeit to a lesser extent than in NREM. Indeed, studies have shown that the phase of ultraslow EEG oscillations during WAKE explains fluctuations in behavioral performance and arousal (Fox et al. 2007 [↗](#), Monto et al. 2008 [↗](#), Sihn and Kim 2022 [↗](#)). In our study, including both NREM and WAKE periods allowed us to additionally show that the temporal organization of activity relative to ultraslow 5-HT oscillations operates according to the same principles in both states, namely by segregating high arousal activity from low arousal internal processing. Rather than a special feature of NREM, the ultraslow 5-HT oscillation appears to be a more fundamental rhythm structuring brain activity. Along these lines, a recent study reported ultraslow oscillations in the firing of medial entorhinal cortex neurons which persisted throughout movement and immobility periods, substates of waking behavior (Gonzalo Congo *et al.* 2024).

In order for the ultraslow oscillation phase to segregate brain activity, as we have observed, the hippocampal network must somehow be able to sense the direction of change of serotonin levels. While single-cell mechanisms related to membrane potential dynamics are typically too fast to explain this calculation, a theoretical work has suggested that feedback circuits can enable such temporal differentiation, also on the slower timescales we observe (Tripp and Eliasmith, 2010 [↗](#)). Beyond the direction of change in serotonin levels, temporal differentiation could also enable the hippocampal network to discern the steeper rising slope versus the flatter descending slope that we observe in the ultraslow 5-HT oscillations (**Figure S2** [↗](#)), which may also be functionally relevant (Cole and Voytek, 2017 [↗](#)). The distinction between the rising and falling phase of ultraslow oscillations is furthermore clearly discernible at the level of unit responses, with many units showing preferences for either half of the ultraslow period (**Figure S6** [↗](#)). Another factor that could help distinguish the rising from the falling phase is the level of other neuromodulators, as it is likely the combination of many neuromodulators at any given time that defines a behavioral substate. Given the finding that ACh and Oxt exhibit ultraslow oscillations with a temporal shift (Zhang et al. 2024 [↗](#)), one could posit that distinct combinations of different levels of neuromodulators could segregate the rising from the falling phase via differential effects of the combination of neuromodulators on the hippocampal network.

In our data, we observed that the rising phase of ultraslow 5-HT oscillations was linked to arousal-associated brain activity. Specifically, long range coherence, which has been shown to correlate with arousal and behavioral performance (Klaver et al. 2023 [↗](#), Parto-Dezfouli et al. 2023 [↗](#)), was seen to peak in the rising phase of ultraslow 5-HT oscillations across a broad range of frequencies. In this way, 5-HT can be observed to gate communication between the cortex and hippocampus, reducing such communication during the ripple-associated falling phase. This gating could serve to reduce sensory input during ‘internal’ hippocampal memory processing, effectively decreasing

potential interference that would disrupt memory consolidation (Logothetis *et al.* 2012 [↗](#), Yang *et al.* 2019 [↗](#)). While the mechanism of 5-HT's effect on long-range neural synchrony is not yet clear, it has been shown that 5-HT can alter sensory gating dependent on communication between the thalamus and hippocampus (Lee *et al.* 2020 [↗](#)). Furthermore, a recent study showed that type 2 dentate spikes (DS2s) in the hippocampus constitute substates of high arousal within immobility periods (Farrell *et al.* 2024 [↗](#)). During DS2s, greater brain-wide activation was observed than during ripples, which mirrors the higher inter-areal coherence and likely higher arousal we observed during the rising phase of ultraslow 5-HT oscillations compared to the falling phase, when ripples preferentially occurred. The role of arousal-associated axo-axonic cells in producing DS2s suggests that these inhibitory cells may be a good target to further examine changes in arousal relative to ultraslow 5-HT oscillations. Indeed, we observed a subset of putative interneurons firing phase-locked to the rising phase (S6), although future work would be required to establish their identity.

Microarousals constitute periods of heightened arousability within NREM which have been hypothesized to maintain a link between the sleeper and the outside world (Halász *et al.* 2004). That 5-HT signaling can affect MA occurrence has been shown in a psilocin study (Thomas *et al.* 2022). Furthermore, given the known association of MAs with increased hippocampal-cortical synchrony (dos Santos Lima *et al.* 2019 [↗](#)), it is not surprising that they, as well as their associated icEMG peaks, like coherence, show a preference for the rising phase of ultraslow 5-HT oscillations. In fact, arousal from NREM sleep was shown to be more likely during NREM microstates with higher inter-cortical coherence (Bastuji *et al.* 2021 [↗](#)). During WAKE, the icEMG signal was locked in the same way to ultraslow 5-HT oscillations, with peaks occurring during the rising phase. This link between serotonin and a correlate of movement is not surprising, as some studies have shown serotonergic control of movement in the hippocampus. Specifically, local infusion of 5-HT into the hippocampus was seen to induce locomotion and serotonergic fibers in CA1 have been shown to activate upon movement initiation (Takahashi *et al.* 2000 [↗](#), Luchetti *et al.* 2020 [↗](#)). Given that the phase of ultraslow serotonin oscillations times icEMG peaks, further investigating the reported link between 5-HT and repetitive movements in the context of 5-HT oscillations could shed more light on the question (Alvarez *et al.* 2022 [↗](#), Jacobs and Fornal 1991 [↗](#)).

According to studies performed to date, increasing 5-HT levels reduces ripple incidence both *in vitro* (ul Haq *et al.* 2016 [↗](#)) and *in vivo* (Wang *et al.* 2015 [↗](#), Shiozaki *et al.* 2023 [↗](#)). Based on these findings, one would expect a negative linear relationship in which ripples occur at the trough of 5-HT fluctuations, as was reported in the case of acetylcholine (Yang *et al.* 2021). In our study, the preference of ripples for the falling phase of ultraslow 5-HT oscillations shows that the dynamics of 5-HT change are more determining for ripple occurrence than the absolute 5-HT level, at this time scale. However, when looking at a longer time scale, namely across states, the relationship between 5-HT level and ripple incidence shows an inverted-U shape with ripples occurring preferentially at the intermediate 5-HT levels observed in NREM (**Figure 6** [↗](#)). This inverted-U dose-response relationship was also found in preliminary hippocampal slice experiments, where the wash-in of low concentration 5-HT increased, and higher concentration 5-HT decreased, ripple incidence (**Figure S1** [↗](#)).

Existing support for the importance of 5-HT release dynamics in consequent brain activity and behavioral outcomes comes from a study showing different behavioral consequences of burst versus tonic 5-HT release (Oikonomou *et al.* 2019 [↗](#)). Given their own findings that burst, but not tonic DRN stimulation induced waking, as well as studies showing that burst-firing of DRN neurons is associated with salient events (Cohen *et al.* 2015 [↗](#), Paquetet *et al.* 2022 [↗](#)), Oikonomou and colleagues posited that 5-HT released in bursts is arousing, whereas tonic release controls slow behavioral state changes, such as increasing sleep drive during wake behavior. Along these lines, the regular increased burst firing observed in a subset of DRN neurons at ultraslow frequencies *in vitro* (Milner *et al.* 2016), likely corresponding to the rising phase of ultraslow 5-HT oscillations in our data, could be seen to signal the regular arousing signals which we observe in

Ripple Incidence by 5-HT Level

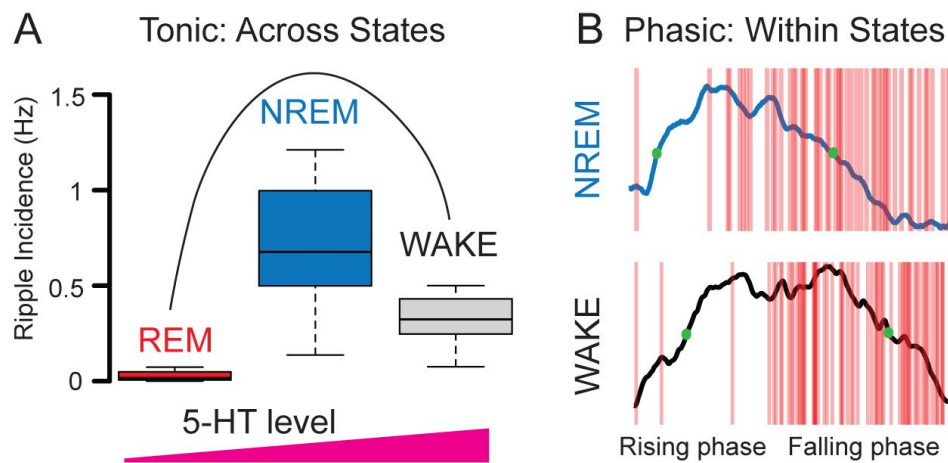


Figure 6.

Relationship between ripple incidence and 5-HT levels depends on time-scale.

A. Ripple incidence by behavioral state shows an inverted-U dose response relationship, with a peak at intermediate 5-HT levels (see **Figure 1F**). **B.** Within states, ripple incidence depends on the phase of the ultraslow 5-HT oscillation. At the same absolute 5-HT level (e.g. green dots), therefore, different ripple incidences are observed.

our ultraslow 5-HT oscillation-defined substates. Ambient 5-HT levels arising from slower changes in tonic state-dependent firing, on the other hand, could dictate the incidence range in which ripples can occur on a slower time-scale, as in our preliminary slice experiments involving the slow wash-in of 5-HT.

A potential mechanism for how different 5-HT release dynamics could differentially affect the hippocampal network at the synaptic level comes from a study on extrasynaptic 5-HT release (Treat and de-Miguel 2012). In this study, high frequency (10-20 Hz), but not low frequency (1 Hz) stimulation was shown to induce extrasynaptic release of 5-HT in the leech Retzius neuron. The effect of such extrasynaptic release would be the targeting of receptors and/or neurons not affected by exclusively synaptic release, thus changing the overall network response to 5-HT. Further support for this idea comes from a recent study showing that during tonic, and low frequency phasic activity of serotonergic neurons, synaptic transmission predominates, while during high frequency and synchronized phasic activity, volume transmission occurs (Zhang et al. 2025 [DOI](#)).

While ripple incidence was biased to the falling phase of ultraslow 5-HT oscillations, higher power ripples were found to cluster around the peak. It follows that, during NREM, the peak of 5-HT oscillations could define a heightened period of ripple propagation to the cortex, which has been shown to be greater in higher power ripples throughout the brain (Nitzan et al. 2020 [DOI](#)), as well as within the hippocampus (de Filippo and Schmitz 2023). In support of this claim, we found cortical power in high frequencies (100-150 Hz.) to be greatest around the peak of hippocampal 5-HT oscillations (Figure S5D [DOI](#)). As hippocampal-cortical interaction during ripples is thought to be a key factor in the consolidation of memory during NREM, the peak of ultraslow 5-HT oscillations could be seen to time memory consolidation itself (Rothschild et al. 2017 [DOI](#), Maingret et al. 2016 [DOI](#)). Further studies are necessary to clarify the relationship between ultraslow 5-HT oscillations, ripple propagation and memory processes.

The two previously mentioned *in vivo* studies showed reduced ripple incidence after *systemically* increasing 5-HT levels, either through intraperitoneal injections of an SSRI, or global activation of median raphe nucleus (MRN) neurons (Wang et al. 2015 [DOI](#), Shiozaki et al. 2023 [DOI](#)). Given the regional specificity of the 5-HT system, such systemic activation has the potential to introduce effects both non-specific to the region of interest, and potentially non-physiological. Systemic administration of a 5-HT₄ receptor agonist, for example, was shown to inhibit locomotion in an open field test, whereas local manipulation of CA1 terminals did not (Teixeira et al. 2018 [DOI](#)). Furthermore, different 5-HT fibers in CA1 were shown to be active during movement initiation and reward (Luchetti et al. 2020 [DOI](#)), indicating that even activating 5-HT fibers within one region at the same time has the potential to activate systems which are not naturally active together. Finally, the mode of systemic release has been shown to make a difference in the resulting behavioral outcome and neural response. In one study, phasic and chronic stimulation of the DRN were shown to inhibit and promote locomotion, respectively (Correia et al. 2017 [DOI](#)). In another study, DRN neurons were shown to have both immediate, *i.e.* phasic, responses to reward and punishment, but also adjust their tonic firing on the time scale of minutes (Cohen et al. 2015 [DOI](#)).

The conclusion of studies showing reduced ripples with increased 5-HT can further be understood in terms of physiological dose. Inhibitory 5-HT_{1a} receptors have been shown to be expressed extrasynaptically in CA1 pyramids (Read et al. 2000). Therefore, one could imagine that after *in vitro* bath application of high concentration 5-HT, or excess stimulation of 5-HT terminals, leading to extrasynaptic release and/or volume transmission of 5-HT, these receptors could be selectively engaged to silence pyramids and shut down the network. In fact, an inverted-U dose response curve with suppression at higher 5-HT levels was found both between 5-HT levels and ripples, in our preliminary *in vitro* wash-in experiments, as well as in a computational study of the effect of 5-HT on spatial working memory in the medial prefrontal cortex (Cano-Colino et al. 2014 [DOI](#)).

In addition to the difficulties involved with typical causal interventions already mentioned, the fact that the levels of different neuromodulators are inter-related and affected by ongoing brain activity makes it very hard to pinpoint ultraslow oscillations of one specific neuromodulator as controlling specific activity patterns, such as ripple timing. While a recent paper purported to show a causative effect of norepinephrine levels on ultraslow oscillations of sigma band power, the fact that optogenetic inhibition of locus coeruleus (LC) cells, but also excitation, only caused a minor reduction of the ultraslow sigma power oscillation suggests that other factors also contribute (Osorio-Ferero *et al.* 2021 [↗](#)). Generally, it is thought that many neuromodulators together determine brain states in a combinatorial manner, and it is probable that the 5-HT oscillations we measure, like the similar oscillations in NE, are one factor among many.

Nevertheless, given the known effects of 5-HT on neurons, it is not unlikely that the 5-HT fluctuations we describe have some impact on the timing of ripples, MAs, hippocampal-cortical coherence, or EMG signals that correlate with either the rising or descending phase. In fact, causal effects of 5-HT on ripple incidence (Wang *et al.* 2015 [↗](#), ul Haq *et al.* 2016 [↗](#) and Shiozaki *et al.* 2023), MA frequency (Thomas *et al.* 2022), sensory gating (Lee *et al.* 2020 [↗](#)), which is subserved by inter-areal coherence (Fisher *et al.* 2020), and movement (Takahashi *et al.* 2000 [↗](#), Alvarez *et al.* 2022 [↗](#), Jacobs *et al.* 1991 and Luchetti *et al.* 2020) have all been shown. Our added findings that serotonin affects ripple incidence in hippocampal slices in a dose-dependent manner (**Figure S1** [↗](#)) further suggests that the relationship between ultraslow 5-HT oscillations and ripples we report may indeed result, at least in part, from a direct effect of serotonin on the hippocampal network.

Whether these ‘causal’ relationships between 5-HT and the different activity measures we describe can be used to support a causal link between ultraslow 5-HT oscillations and the correlated activity we report remains an open question. To that point, some studies have described changes in ultraslow oscillations due to manipulation of serotonin signaling. Specifically, reduction of 5-HT_{1a} receptors in the dentate gyrus was recently shown to reduce the power of ultraslow oscillations of calcium activity in the same region (Turi *et al.* 2024 [↗](#)). Furthermore, psilocin, which largely acts on the 5-HT_{2a} receptor, decreased NREM episode length from around 100 s to around 60 s, and increased the frequency of brief awakenings (Thomas *et al.* 2022). While ultraslow oscillations were not explicitly measured in this study, the change in the rhythmic pattern of NREM sleep episodes and brief awakenings, or microarousals, suggests an effect of psilocin on ultraslow oscillations during NREM. Although these studies do not necessarily point to an exclusive role for 5-HT in controlling ultraslow oscillations of different brain activity patterns, they show that changes in 5-HT can contribute to changes in brain activity at ultraslow frequencies.

These methodological considerations highlight the benefit of the correlative approach adopted here, measuring local 5-HT levels and brain activity simultaneously. While causal relationships cannot be determined from our *in vivo* strategy, the relationship observed between 5-HT levels and ripples can be used to inform future causal studies in a data-driven way. For example, the findings highlight the importance of having a detailed look at the relationship between different 5-HT release dynamics and hippocampal cell and network responses. Furthermore, modulating the frequency or strength of the ultraslow 5-HT oscillations, as done in (Osorio-Ferero *et al.* 2021 [↗](#)) for spindle oscillations and noradrenaline, could provide insight into how 5-HT tone and phasic release modulate ripples in a realistic setting.

Materials and Methods

Animals

All experimental procedures were performed following the Guide for Animal Care and Use of Laboratory Animals. Male C57BL6/J mice (Jackson Laboratory) between 2-6 months of age were used for experiments. The mice were housed in groups of 2-5 animals prior to surgery, and singly after surgery, in a reverse 12/12 hour light-dark cycle (lights on 10 p.m. to 10 a.m.) with ad libitum access to food and water.

Surgery

For viral injection, as well as for the implantation of optic fiber cannulae and silicon probes, mice were anesthetized with isoflurane (4%) and placed in a stereotactic frame (Kopf Instruments). Body temperature was maintained at 38° C by a heating pad (Harvard Instruments). The isoflurane level was slowly reduced to 1-2 % to maintain anesthesia throughout the surgery. Mice were injected with ketoprofen (10 mg/kg, s.c.). Hair was removed with a depilatory cream, the scalp was cleaned with ethanol and iodine solutions, and the skull exposed.

Viral Injection and Optic Fiber Implantation

A craniotomy was performed by drilling a small hole above the right dorsal CA1 (AP -2.3 mm, ML -2.00 mm). A glass injection micropipette with a 100 µm tip was pulled, filled with mineral oil, and connected to a Hamilton syringe attached to a microsyringe pump (KD Scientific, Legato 111). 250 nl of AAV9-hSyn-5HT3.0 (WZ Biosciences) was injected at a rate of 100 nl/min and a depth of 1.3 mm below the dura. After injection, the pipette was left in place for 5 minutes before slowly bringing it up out of the brain over the course of 20-30 minutes. Saline was administered to the craniotomy site to keep the tissue moist throughout the procedure.

An optic fiber (Thorlabs CFML15L05) was then implanted above the injection site. Only fibers with >80 % transmission efficacy were used. The optic fiber was secured with C&B Metabond (Parkell). Dental cement was applied to exposed areas of the skull. Mice were kept in their home cages for 3 weeks to allow recovery from surgery and expression of the virus.

Silicon Probe Implantation and Electrophysiological Recordings

Three weeks after viral injection and optic fiber implantation, a second surgery was performed to implant a silicon probe (NeuroNexus, 64-channel, edge, 1 or 4 shanks), mounted onto a microdrive (NeuroNexus, dDrive) into the left dorsal CA1. To this end, anesthesia was induced, as previously described, and a mouse cap with copper mesh (3DNeuro) was cemented to the skull. A second craniotomy was then performed over the right dorsal CA1 (AP -2.3 mm, ML +2.00 mm). The probe was slowly lowered and secured with C&B Metabond at a depth of 0.8 mm below the dura. The exposed brain was covered with a mixture of heated bone wax and mineral oil. A grounding screw was placed over the cerebellum and soldered to the ground electrode on the probe and the mouse cap. The mouse was allowed to recover for five days, at which point the probe was further lowered until the prominent spikes and sharp wave ripples characteristic of the CA1 pyramidal layer were observed. Data was recorded with an Open Ephys system at 20 kHz.

Fiber photometry

Fiber photometry was performed as described in (Krok et al. 2023 [link](#)). Briefly, a fiber-coupled 470 nm LED (Thorlabs) was used to send excitation light in continuous wave mode through a fiber optic patch cord (Doric) to the mouse's optic fiber implant via a fluorescence mini-cube (Doric). Emitted light traveled back through the same optic fiber patch cord to the mini-cube, and was collected by a photoreceiver (Newport, DC mode). Signal collected by the photoreceiver was digitized at 5 kHz with a National Instruments acquisition board (NI BNC-2090A) and analyzed using Wavesurfer (Janelia).

Preprocessing of fiber photometry data was performed as described by Thomas Akam (Github, Thomas Akam, photometry_preprocessing: https://github.com/ThomasAkam/photometry_preprocessing/tree/master [↗](#)).

Namely, raw fiber photometry data was first downsampled to 1250 Hz for comparison with electrophysiology data, and low pass filtered at 20 Hz. using MATLAB's (R2021a) lowpass function. Next slow drift in signal due to photobleaching was corrected by fitting a second-order exponential function. Finally, in order to compare photometry data across sessions and mice, the signals were z-scored.

Fluoxetine injections

In order to show that the GRAB5-HT3.0 sensor used is responsive to changes in local 5-HT levels, 10 mg/kg fluoxetine-hydrochloride (Sigma-Aldrich) dissolved in saline or saline only (control) was administered intraperitoneally after a 20 min. baseline.

Dual fiber photometry and silicon probe recordings

Contralateral simultaneous recordings were chosen over ipsilateral due to the size of the optic fibers and fragility of the silicon probes, which prevented their implantation in close proximity. In addition to its adoption in a recent dual-recording study (Yang *et al.* 2024), this contralateral recording scheme can be justified due to the simultaneous occurrence of ripples, a major electrophysiological read-out in the current study, as well as dentate spikes across hemispheres (Chrobak and Buzsáki 1996 [↗](#), Buzsáki 2003, Jarrell *et al.* 2024), bilateral synchrony of ultraslow EEG oscillations (Liu *et al.* 2010 [↗](#)), as well as the bilateral symmetry of the 5-HT system (Ren *et al.* 2018 [↗](#)).

Fiber photometry and electrophysiological data were simultaneously recorded from double-implanted mice in their home cages for 2-3 hour sessions, containing both wake and sleep periods. Synchronization of photometry and electrophysiological data was performed by triggering recording onset with an Arduino pulse.

CNN for ripple detection

The custom CNN model used for ripple detection was inspired by the approach of (Navas-Olive *et al.* 2022 [↗](#)) and can be found on GitHub (Cooper 2024). In their work, ripple detection was reframed from 1D thresholding of spectral features over time, to an image recognition problem, where the image consists of segments of LFP data from multiple channels containing ripples. Detection thus takes into account the classic laminar pattern of ripples, which is useful for distinguishing them from other fast oscillations or noise. Furthermore, this approach is unbiased in the sense that it doesn't rely upon a strict limitation of ripple features and, furthermore, has been shown to perform more consistently with data from different experimental sessions than standard filter-based detection (Navas-Olive *et al.* 2024 [↗](#)).

Data preparation

In constructing our model, which is structurally simpler than that proposed in (Navas-Olive *et al.* 2022 [↗](#)). Inputs to the network were prepared as follows: 4 neighboring channels in CA1 showing ripples were chosen. With 50 μ m spacing between electrode sites, 4 channels showing ripple oscillations displayed characteristic amplitude changes that were key to distinguishing them from movement artifacts, fast gamma oscillations or other false positives often detected as ripples by traditional ripple detection algorithms. Additionally, four channels were chosen from the neocortex to increase the network's ability to rule out movement artifacts or other noise which propagates uniformly across channels. Data from the 8 channels was then z-scored and segmented into 400 ms chunks and these 8 channel x 400 ms chunks were fed in as 2D inputs to the network.

Ripple Annotation and Training Data

To prepare a training set, ripple start and stop times were labeled for 5,000 ripples occurring in data from three different mice. A key criterion for distinguishing ripples from non-ripples was signal modulation across hippocampal and cortical channels. Fast and transient oscillations in which the amplitude varied according to location relative to the center of the CA1 pyramidal layer were considered ripples, whereas oscillations in which the amplitude appeared constant across hippocampal or hippocampal and cortical channels were excluded. 400 ms data segments centered around the labeled ripples were extracted for the 8 input channels (4 hippocampal and 4 cortical). A label trace (1 x 400 ms) was generated for each segment in which time outside of ripples was taken to be 0, and time during ripples, 1. For training, 5,000 negative examples, i.e. data segments including no ripples, were also included. The timing of these segments were chosen at random such that there was no overlap between them and the ripple-containing segments, and were taken from the same 8 channels and three mice used for ripple-containing segments. Label traces were generated for each negative example consisting of a 1 x 400 ms trace of zeroes. Training data was split into a training set (80%) and a testing set (20%).

CNN Architecture

A custom convolutional neural network (CNN) was built using Python 3.9.12 and the following key python packages:

tensorflow 2.9.1

keras 2.9.0

numpy 1.23.0

scipy 1.8.1

The model was built within a custom model class called RipNet. RipNet consisted of four convolution blocks, with each block consisting of a 2D convolutional layer (Conv2D), followed by a Rectified Linear Unit (ReLU) activation layer and a Batch Normalization layer. Stride length was (2,2) for all blocks and kernel size was (1,1) for the first block, and (3,3) for the subsequent three blocks. The convolution blocks were followed by a Dropout layer (0.25), a Dense layer, a Batch Normalization layer, and a second Dropout layer (0.4). A sigmoid activation function on the final Dense layer provided the prediction trace, which gave the likelihood of ripple occurring as a number between 0 and 1, for the input trace provided. Ripples were detected when the prediction trace exceeded an empirically-determined threshold of 0.2. Ripple peak times were determined by taking the peak of the envelope of the ripple-band (120-250 Hz.) bandpass-filtered signal.

Network architectures, i.e. the number of convolutional blocks, and hyperparameters, including the optimizer, learning rate and regularization, were tuned during training. After training, the best-performing model consisted of four convolutional blocks, an Adam optimizer with a learning rate of 1e-4 and a decay rate of 1e-4, as well as L2 regularization (0.001) to prevent overfitting. Mean-squared error (MSE) was used as a loss function to compare the predicted trace to the ground truth trace.

Model performance

Model performance was evaluated based on ripples detected in two hours of data across two mice, which was not included in the training dataset. In this testing set, ripples were annotated manually and compared to model predictions. True positives (TPs) occurred when manually labeled ripples were also predicted by the model. False negatives (FNs) were marked where

ripples were annotated, but not predicted by the model. False positives (FPs) encompassed ripples predicted by the model and not labeled manually, and upon second inspection, not considered ripples. Based on these metrics, the F1 value, which represents the harmonic mean of precision ($TP / (TP + FP)$) and recall ($TP / (TP + FN)$), was calculated as a measure of model performance. The F1 value was found to be 81.5% for the testing data set. Of note, this F1 value is higher than that reported for both the standard Butterworth filter method at optimized performance and the aforementioned previously published CNN (Navas-Olive et al. 2022 [DOI](#)), with F1 values of 68% and 65%, respectively. To ensure that our model did not systematically exclude ripples with certain features in a way that would influence their overall relationship to ultraslow 5-HT oscillations, we took a closer look at the ripples missed by the model. Specifically, we compared ripple power in these missed ripples to power in correctly identified ripples (**Figure S7A-B** [DOI](#)). Ultimately, we showed that, despite the model performing slightly better during the falling phase, as indicated by higher recall, inclusion of these missed ripples did not affect the fundamental relationship between ripples and ultraslow 5-HT oscillations (**Figure S7C** [DOI](#)).

State scoring

Behavioral states were designated as WAKE, NREM, REM or MA according to output of the SleepScoreMaster function from the Buzsaki lab code repository (<https://github.com/buzsakilab/buzcode> [DOI](#)). The methodology for SleepScoreMaster's sleep score detection is described in (Watson et al. 2016 [DOI](#)). Briefly, the LFP power at low frequencies (< 25 Hz.) was first used to distinguish NREM from 'other' states. Next, the 'other' states were assigned labels according to the narrow theta-band ratio (5-10 Hz. / 2-6 Hz.) and the intracranial EMG (icEMG), with high theta-band ratio and low EMG corresponding to REM states, and remaining states being classified as WAKE (>40 s) or MA (<40 s). Detected states were then reviewed manually by the experimenter. The icEMG was calculated as part of the SleepScoreMaster function as described in Watson et al. 2016 [DOI](#). Briefly, the correlations between the signals from spatially distal pairs of channels, bandpass filtered from 300-600 Hz. was computed in 0.5 s bins. The average of these correlations was then taken as the icEMG. Notably, the icEMG, as calculated here, and traditional EMG signals show a high correlation (Schomburg et al. 2014 [DOI](#)), and discrepancies between the two measures were shown to reflect, e.g., cases where the animal's face moved with no overall head displacement, such as occurs during chewing (Watson et al. 2016 [DOI](#)).

LFP Analysis

LFP analysis was performed using custom MATLAB code. Time-frequency power spectra were generated using the Stockwell Transform (Stockwell et al. 1996 [DOI](#)) with the function `st` (Sundar 2015 [DOI](#)). Time-frequency power spectra were normalized to 0-1 for visualization. Phase angles were calculated using the Hilbert transform of the bandpass-filtered signal (0.01-0.06 Hz.). Magnitude-squared wavelet coherence was calculated using MATLAB's `wcoherence` function.

Histology

Transcardiac perfusion was performed after deep anesthesia with isoflurane with 4% paraformaldehyde (PFA) in 0.1 M sodium phosphate buffer. Brains were kept in PFA overnight, then sliced into 100 μ m coronal sections with a vibratome (Leica). Slices were mounted using Fluoroshield with DAPI (Sigma) and endogenous fluorescence from the GRAB5-HT3.0 sensor was imaged with an Olympus VS120 slide scanning microscope. From these images, the expression of the GRAB5-HT3.0 sensor in the dorsal CA1 was verified.

Statistics

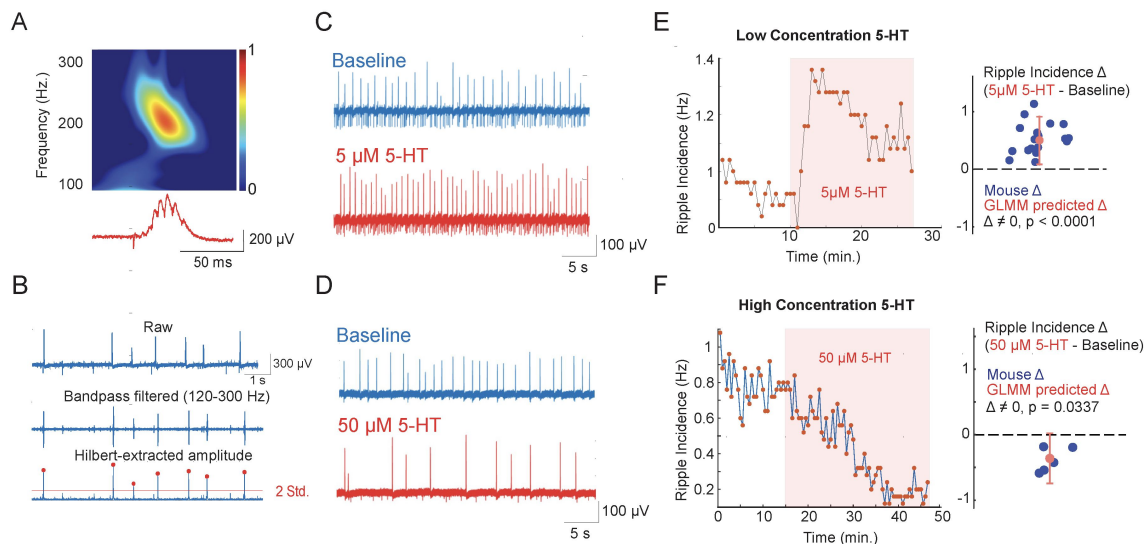
All plots with error bars or bounded lines reflect the mean across sessions $\pm$ standard error of the mean. Statistics were performed in RStudio, using R version 4.3.2. As the data for all experiments is hierarchical, it is necessary to account for inter-mouse and inter-session variation (Yu et al. 2022 [DOI](#)). To this end, we fit general linear mixed models (GLMMs), including a fixed effect term of

state (NREM, WAKE) and random effects terms for mouse and session ID. Unless otherwise specified, p values were obtained with *emmeans*, which was used for post-hoc testing of fitted models (Lenth 2024 [↗](#)) as well as a Bonferroni adjustment for multiple comparisons. For **Figure 1F** [↗](#), the *blme* package was used with a gaussian link function (Chung et al. 2013 [↗](#)). For **Figure 3F2,4** [↗](#) and **Figure 4C** [↗](#) a binomial link function was used with the *lme4* package (Bates et al. 2015 [↗](#)). Significant phase preference for the reference state (NREM) was inferred from the significance of the intercept. To determine whether the non-reference state (WAKE) also exhibited a significant bias, we calculated its log-odds by adding the intercept and the state coefficient. The corresponding z-score was computed, and the p-value was then obtained from a two-tailed normal distribution. As MAs occur only during NREM, state was no longer a fixed effect, and significance was inferred from the significance of the model's intercept. For **Figure S2B** [↗](#), *glmer()* from *lme4* was used with no link function (Bates et al. 2015 [↗](#)). For **Figure 1H** [↗](#), **3H** [↗](#), **S1** [↗](#) and **S5** [↗](#) a gamma link function was used with the *glmmTMB* package (Brooks et al. 2017). For **Figures 4G** [↗](#) and **5D** [↗](#), a beta distribution link function was used with the *glmmTMB* package. For **Figure S2D** [↗](#), the *lme4* package was used (Bates et al. 2015 [↗](#)). Ratios were log-transformed such that the significance of the intercept reflected the difference of non-transformed ratios from 1. For **Figure 1I** [↗](#), a Wilcoxon ranked-sum test was used on the mean 5-HT levels in fluoxetine and saline-injected mice in the 20 minute period starting 20 minutes after the injection.

Data availability

Data required to reproduce findings can be found on FigShare: <https://figshare.com/account/home#/projects/204408> [↗](#).

Supplementary Figures



Supplementary Figure 1.

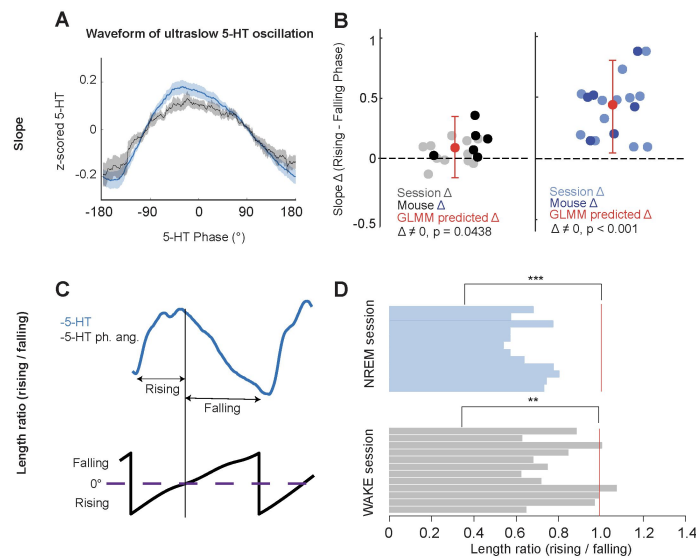
Serotonin modulates ripple incidence *in vitro*.

A. Spectrogram of an example *in vitro* ripple, 0-1 normalized. **B.** Ripple detection *in vitro*. Given the absence of non-ripple fast oscillations and movement noise *in vitro*, ripples were detected in the standard way; namely, the raw LFP signal was bandpass filtered in the ripple range (120-300 Hz.), the amplitude of activity in this frequency band was extracted with the Hilbert transform, and finally ripples were detected as peaks greater than a threshold of two standard deviations and a minimum duration of 20 ms. **C.** Example traces from the baseline (blue) and low concentration 5-HT wash-in period (red) of an example recording. **D.** Example traces from the baseline (blue) and high concentration 5-HT wash-in period (red) of an example recording. **E.** Left: Ripple incidence over time in an example wash-in experiment. 5 μ M 5-HT was added to the circulating ACSF after a 10-minute baseline. Right: Mean 5 μ M 5-HT wash-in ripple incidence - mean baseline ripple incidence, plotted by mouse. Red point with error bar indicates predicted difference and confidence interval after fitting a GLMM to the data. P-value shown was derived from a post-hoc multiple comparisons test on the fitted model. **F.** Left: Ripple incidence over time in an example wash-in experiment. 50 μ M 5-HT was added to the circulating ACSF after a 10-minute baseline. Right: Mean 50 μ M 5-HT wash-in ripple incidence - mean baseline ripple incidence, plotted by mouse ($n = 20$ slices from 5 mice). Red point with error bar indicates predicted difference and confidence interval after fitting a GLMM to the data. P-value shown was derived from a post-hoc multiple comparisons test on the fitted model.

Supplementary Figure 2.

Waveform of ultraslow 5-HT oscillation is asymmetric.

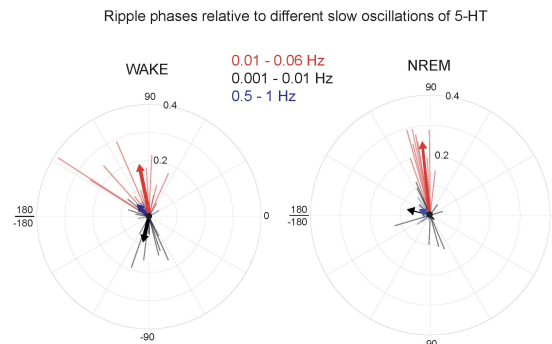
A. Mean 5-HT waveforms for one period of the ultraslow 5-HT oscillation during WAKE (black) and NREM (blue). **B.** Mean rising phase 5-HT slope - mean falling phase 5-HT slope, plotted by session and mouse for WAKE (left) and NREM (right) data. Red point with error bar indicates predicted difference and confidence interval after fitting a GLMM to the data. P-values shown were derived from a post-hoc multiple comparisons test on the fitted model. **C.** Example 5-HT trace showing one period of the ultraslow oscillation with corresponding phase angles below to illustrate the calculation of the length ratio (rising phase length / falling phase length). **D.** Length ratio per session in NREM (top) and WAKE (bottom). Line at 1 indicates equal rising and falling length. Significant difference from equal rising and falling lengths (ratio = 1) was tested after fitting a GLMM to the log-transformed data (NREM: $p < 0.0001$, WAKE: $p = 0.003$).

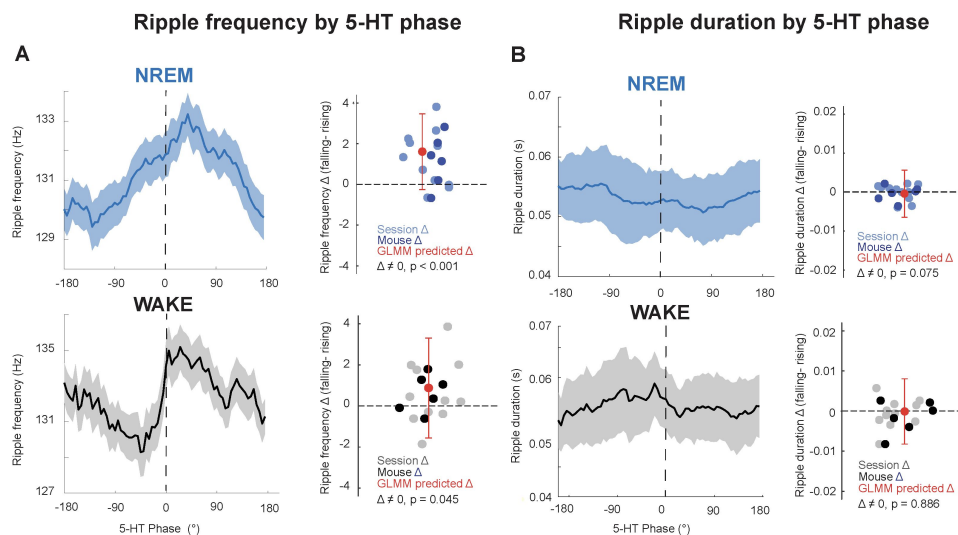


Supplementary Figure 3.

Ripples couple most strongly to ultraslow (0.01 - 0.06 Hz.) oscillations of serotonin.

Individual lines represent session-level mean phase vectors of ripples relative to different slow oscillations of 5-HT (red: 0.01-0.06 Hz., black: 0.001-0.01 Hz., blue: 0.5-1 Hz.) for WAKE (left) and NREM (right). Thicker lines with arrows represent the grand mean vectors (mean of session-level means) for each frequency band.

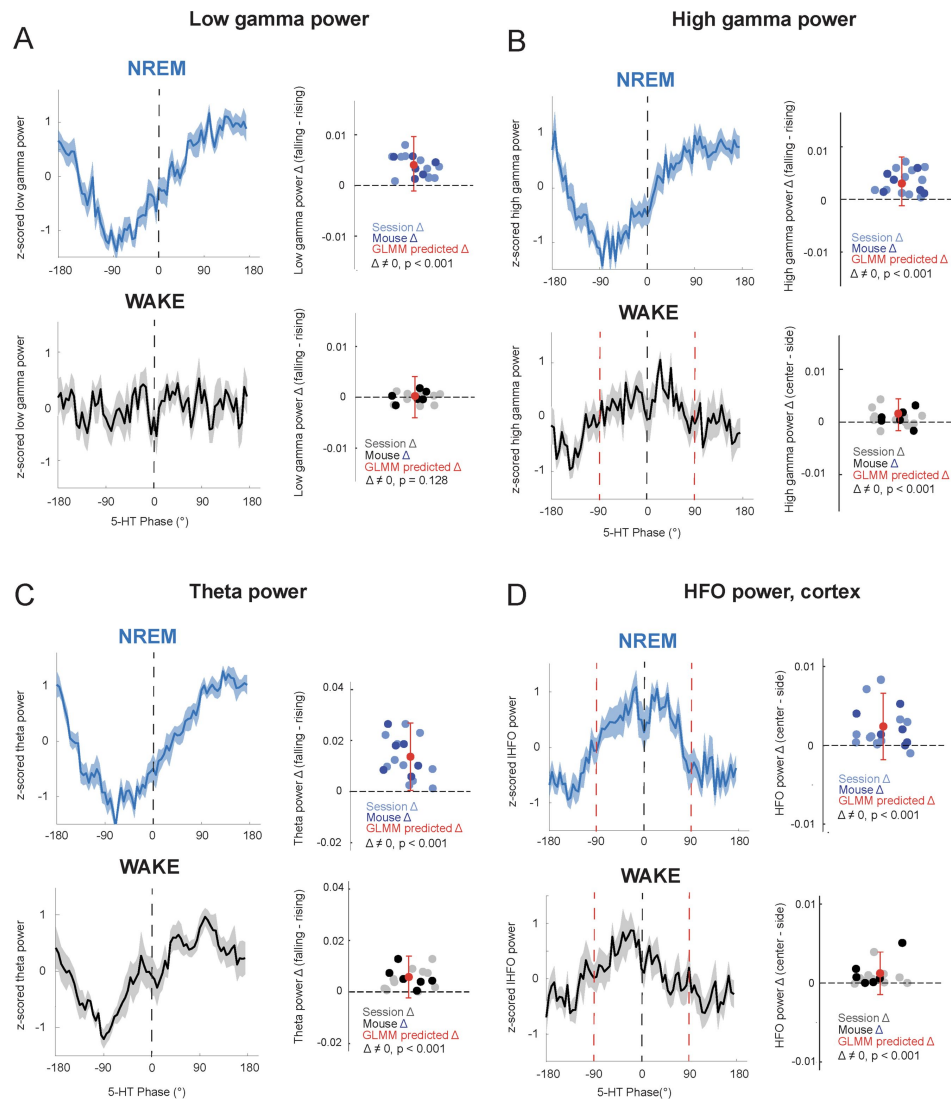




Supplementary Figure 4.

Ripple frequency, but not duration, is shaped by ultraslow 5-HT oscillations.

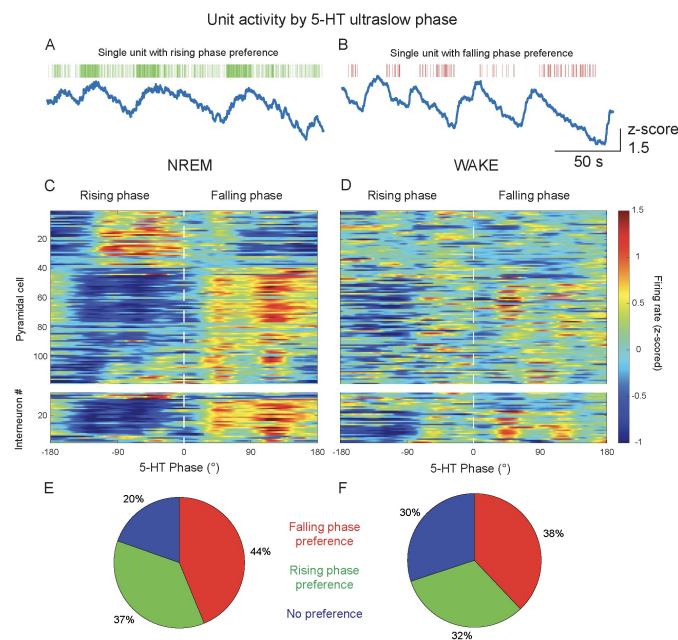
A. Mean ripple frequency by phase of ultraslow 5-HT oscillation in NREM (top) and WAKE (bottom). **B.** Mean falling phase ripple frequency - mean rising phase ripple frequency, plotted by session and mouse. Red point with error bar indicates predicted difference and confidence interval after fitting a GLMM to the data. P-values shown were derived from a post-hoc multiple comparisons test on the fitted model. **C.** Mean ripple duration by phase of ultraslow 5-HT oscillation in NREM (top) and WAKE (bottom). **D.** Mean falling phase ripple duration - mean rising phase ripple duration, plotted by session and mouse. Red point with error bar indicates predicted difference and confidence interval after fitting a GLMM to the data. P-values shown were derived from a post-hoc multiple comparisons test on the fitted model.



Supplementary Figure 5.

Power in different frequency bands is timed by ultraslow 5-HT oscillations.

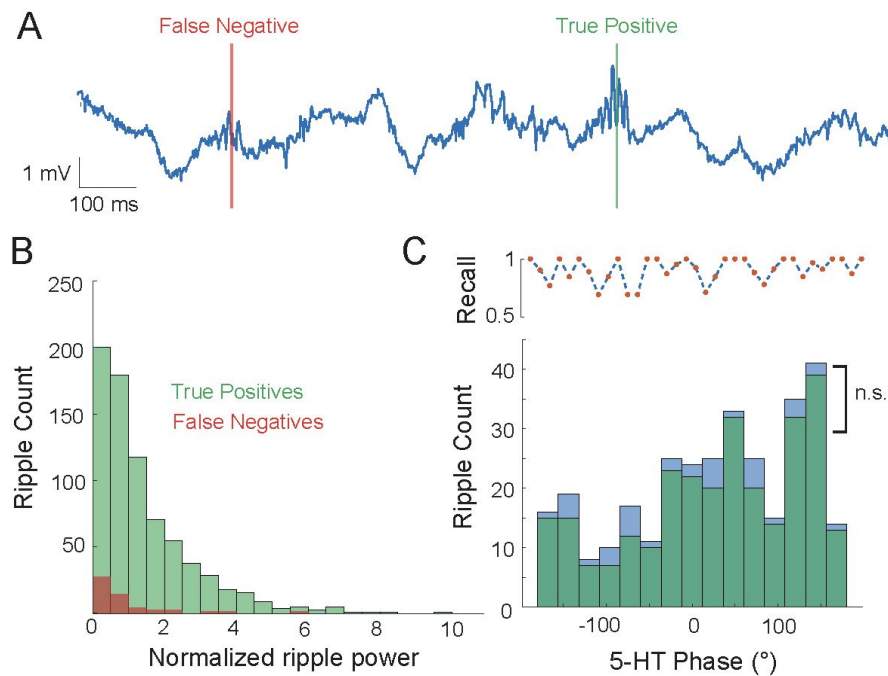
A-D. Left columns: Mean power by phase of ultraslow 5-HT oscillation in NREM (top) and WAKE (bottom) for **A.** low gamma (30 - 55 Hz.), **B.** high gamma (60-90 Hz.), **C.** theta (8-12 Hz.), and **D.** cortical high frequency oscillation (HFO, 100-150 Hz.) **A-D.** Right columns: Mean falling phase low gamma power - mean rising phase low gamma power, plotted by session and mouse during NREM (top) and WAKE (bottom). Red point with error bar indicates predicted difference and confidence interval after fitting a GLMM to the data. P-values shown were derived from a post-hoc multiple comparisons test on the fitted model.



Supplementary Figure 6.

Units show preference for different phases of ultraslow serotonin oscillations.

A. Example spiking activity (green) of a neuron relative to the ultraslow serotonin oscillation with a preference for the rising phase. **B.** Example spiking activity (red) of a neuron relative to the ultraslow serotonin oscillation with a preference for the falling phase. **C.** Mean z-scored firing rates of pyramidal cells and interneurons by ultraslow serotonin oscillation phase in NREM. **D.** Same data as in **C** in the WAKE state. **E.** Percentage of units showing falling phase preference (red), rising phase preference (green), and no preference (blue) as per results of a circular V test ($p < 0.05$) in NREM. **F.** Same data as **E** in the WAKE state. ($n=3$ mice, 1 session per mouse, with 64, 59 and 30 extracted units, respectively)



Supplementary Figure 7.

Ripples missed by model do not affect ripple distribution along ultraslow 5-HT oscillations.

A. Example trace showing a correctly detected ripple (green) and a ripple missed by the model (red). **B.** Ripple power of correctly detected ripples (true positives, green) and missed ripples (false negatives, red). Model performance was evaluated manually in three mice with a total of 772 true positives, and 52 false negatives. **C.** Top: Recall (true positives / true positives + false negatives) by phase of ultraslow 5-HT oscillation. Bottom: Histogram showing true positives by 5-HT ultraslow phase angle (green) and true ripples (true positives + false negatives, blue). No statistical difference was found between the two distributions (Mardia-Watson-Wheeler test, $p = 0.9375$).

Acknowledgements

We would like to thank the Buzsàki lab for providing the virus expressing the GRAB5-HT3.0 sensor used in this study, and Dr. Yiyao Zhang for offering some initial training and input on fiber photometry. Additionally, we are grateful to the members of the Tritsch lab for their insightful discussions, sharing lab equipment, and assistance with fiber photometry, especially James Taniguchi and Tony Garcia.’

Additional information

Funding

This study was supported by the German Research Foundation (Deutsche Forschungsgemeinschaft (DFG), project 184695641 – SFB 958, project 327654276 – SFB 1315, project 415914819 – FOR 3004, project 431572356, and under Germany’s Excellence Strategy – Exc-2049-390688087), by the Federal Ministry of Education and Research (BMBF, SmartAge-project 01GQ1420B) and by the European Research Council (ERC) under the Europeans Union’s Horizon 2020 research and innovation program (grant agreement No. 810580).

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Analysis and interpretation of data: C.C., D.P.

Drafting the article: C.C.

Contributed to new analytic tools or reagents: D.P.,J.S.,J.T.,N.T.

Revision of the article: all authors

Note

This reviewed preprint has been updated to correct the author details.

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Reviewer #1 (Public review):

Summary:

In this work, authors recorded the dynamics of the 5-HT with fiber photometry from CA1 in one hemisphere and LFP from CA1 in the other hemisphere. They have observed an ultra-slow oscillation in the 5-HT signal both during wakefulness and NREM sleep. The authors have studied different phases of the ultra-slow oscillation to examine the potential difference in the occurrence of some behavioral state-related physiological phenomena (hippocampal ripples, EMG, and inter-area coherence).

Strengths:

The relation between the falling/rising phase of the ultra-slow oscillation and the ripples is sufficiently shown. There are some minor concerns about the observed relations that should be addressed with some further analysis.

Systematic observations have started to establish a strong relation between the dynamics of neural activity across the brain and measures of behavioral arousal. Such relations span a wide range of temporal scales that are heavily inter-related. Ultra-slow time scales are specifically understudied due to technical limitations and neuromodulatory systems are the strongest mechanistic candidates for controlling/modulating the neural dynamics at these time scales. The hypothesis of the relation between a specific time scale and one certain neuromodulator (5-HT in this manuscript) could have a significant impact on the understanding of the hierarchy in the temporal scales of neural activity.

Weaknesses:

weaknesses appropriately addressed by reviewers in the current version

<https://doi.org/10.7554/eLife.101105.2.sa3>

Reviewer #2 (Public review):**Summary:**

In their study, Cooper et al. investigated the spontaneous fluctuations in extracellular 5-HT release in the CA1 region of the hippocampus using GRAB5-HT3.0. Their findings revealed the presence of ultra-low frequency (less than 0.05 Hz) oscillations in 5-HT levels during both NREM sleep and wakefulness. The phase of these 5-HT oscillations was found to be related to the timing of hippocampal ripples, microarousals, electromyogram (EMG) activity, and hippocampal-cortical coherence. In particular, ripples were observed to occur with greater frequency during the descending phase of 5-HT oscillations, and stronger ripples were noted to occur in proximity to the 5-HT peak during NREM. Microarousal and EMG peaks occurred with greater frequency during the ascending phase of 5-HT oscillations. Additionally, the strongest coherence between the hippocampus and cortex was observed during the ascending phase of 5-HT oscillations. These patterns were observed in both NREM sleep and the awake state, with a greater prevalence in NREM. The authors posit that 5-HT oscillations may temporally segregate internal processing (e.g., memory consolidation) and responsiveness to external stimuli in the brain.

Strengths:

The findings of this research are novel and intriguing. Slow brain oscillations lasting tens of seconds have been suggested to exist, but to my knowledge they have never been analyzed in such a clear way. Furthermore, although it is likely that ultra-slow neuromodulator oscillations exist, this is the first report of such oscillations, and the greatest strength of this study is that it has clarified this phenomenon both statistically and phenomenologically.

Weaknesses:

As with any paper, this one has some limitations. While there is no particular need to pursue them, I will describe ten of them below, including future directions:

Contralateral recordings: 5-HT levels and electrophysiological recordings were obtained from opposite hemispheres due to technical limitations. Ipsilateral simultaneous recordings may show more direct relationships.

Sample size: The number of mice used in the experiments is relatively small (n=6). Validation with a larger sample size would be desirable.

Lack of causality: The observed associations show correlations, not direct causal relationships, between 5-HT oscillations and neural activity patterns.

Limited behavioral states: The study focuses primarily on sleep and quiet wakefulness. Investigation of 5-HT oscillations during a wider range of behavioral states (e.g., exploratory behavior, learning tasks) may provide a more complete understanding.

Generalizability to other brain regions: The study focuses on the CA1 region of the hippocampus. It's unclear whether similar 5-HT oscillation patterns exist in other brain regions.

Long-term effects not assessed: Long-term effects of ultra-low 5-HT oscillations (e.g., on memory consolidation or learning) were not assessed.

Possible species differences: It's uncertain whether the findings in mice apply to other mammals, including humans.

Technical limitations: The temporal resolution and sensitivity of the GRAB5-HT3.0 sensor may not capture faster 5-HT dynamics.

Interactions with other neuromodulators: The study does not explore interactions with other neuromodulators (e.g., norepinephrine, acetylcholine) or their potential ultraslow oscillations.

Limited exploration of functional significance: While the study suggests a potential role for 5-HT oscillations in memory consolidation and arousal, direct tests of these functional implications are not included.

<https://doi.org/10.7554/eLife.101105.2.sa2>

Reviewer #3 (Public review):

Summary:

Activity of serotonin (5-HT) releasing neurons as well as 5-HT levels in brain structures targeted by serotonergic axons are known to fluctuate substantially across the animal's sleep/wake cycle, with high 5-HT during wakefulness (WAKE), intermediate 5-HT levels during non-REM sleep (NREM) and very low 5-HT levels during REM sleep. Recent studies have shown that during NREM, activity of 5-HT neurons in raphe nuclei oscillates at very low frequencies (0.01 - 0.05 Hz) and this ultraslow oscillation is negatively coupled to broadband EEG power. However, how exactly this 5-HT oscillation affects neural activity in downstream structures is unclear.

The present study addresses this gap by replicating the observation of the ultraslow oscillation in the 5-HT system, and further observing that hippocampal sharp wave-ripples (SWRs), biomarkers of offline memory processing, occur preferentially in barrages on the falling phase of the 5-HT oscillation during both wakefulness and NREM sleep. In contrast, the study found that the raising phase of the 5-HT oscillation is associated with microarousals during NREM and increased muscular activity during WAKE. Finally, the raising 5-HT phase was also found to be associated with increased synchrony between the hippocampus and neocortex.

In vivo findings are further supported by an ex vivo demonstration of dose-dependent serotonergic SWR modulation, lends support to the potential causal relationship between 5-HT slow oscillation and hippocampal dynamics.

Overall, the study constitutes a valuable contribution to the field by reporting a close association between, on one hand, raising 5-HT and arousal and, on the other hand, falling 5-HT and offline memory processes.

Strengths:

The study makes a compelling use of the state-of-the art methodology to address its aims: the genetically encoded 5-HT sensor used in the study is ideal for capturing the ultraslow 5-HT dynamics and the novel detection method for SWRs outperforms current state-of-the-art algorithms and will be useful to many scientists in the field. Explicit validation of both of these methods is a particular strength of this study.

The analytical methods used in the article are appropriate and are convincingly applied, the use of a general linear mixed model for statistical analysis is a particularly welcome choice as it guards against pseudoreplication while preserving statistical power.

Pharmacological demonstration of serotonergic SWR modulation in brain slices adds further weight to the possible direct role of 5-HT in hippocampal dynamics in vivo.

Overall, the manuscript makes a strong case for distinct sub-states across WAKE and NREM, associated with different phases of the 5-HT oscillation.

Weaknesses:

All in vivo evidence presented in the study is correlational, although the ex vivo results do suggest a possibility of a causal relationship between 5HT levels and hippocampal dynamics in the intact brain.

<https://doi.org/10.7554/eLife.101105.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary

In this work, the authors recorded the dynamics of the 5-HT with fiber photometry from CA1 in one hemisphere and LFP from CA1 in the other hemisphere. They observed an ultra-slow oscillation in the 5-HT signal during both wakefulness and NREM sleep. The authors have studied different phases of the ultra-slow oscillation to examine the potential difference in the occurrence of some behavioral state-related physiological phenomena (hippocampal ripples, EMG, and inter-area coherence).

Strengths

The relation between the falling/rising phase of the ultra-slow oscillation and the ripples is sufficiently shown. There are some minor concerns about the observed relations that should be addressed with some further analysis.

Systematic observations have started to establish a strong relation between the dynamics of neural activity across the brain and measures of behavioral arousal. Such relations span a wide range of temporal scales that are heavily inter-related. Ultra-slow time-scales are specifically under-studied due to technical limitations and neuromodulatory systems are the strongest mechanistic candidates for controlling/modulating the neural dynamics at these time-scales. The hypothesis of the relation between a specific time-scale and one certain neuromodulator (5-HT in this manuscript) could have a significant impact on the understanding of the hierarchy in the temporal scales of neural activity.

Weaknesses:

One major caveat of the study is that different neuromodulators are strongly correlated across all time scales and related to this, the authors need to discuss this point further and provide more evidence from the literature (if any) that suggests similar ultra-slow oscillations are weaker or lack from similar signals recorded for other neuromodulators such as Ach and NA.

The reviewer is correct to point out that the levels of different neuromodulators are often correlated. For example, most monoaminergic neurons, including serotonergic neurons of the raphe nuclei, show similar firing rates across behavioral states, firing most during wake behavior, less during NREM, and ceasing firing during 'paradoxical sleep' or REM (Eban-

Rothschild et al 2018). Notably, other neuromodulators, such as acetylcholine (ACh), show the opposite pattern across states, with highest levels observed during REM, an intermediate level during wake behavior, and the lowest level during NREM (Vazquez et al. 2001). Despite these differences, ultraslow oscillations of both monoaminergic and non-monoaminergic neuromodulators, have been described, albeit only during NREM sleep (Zhang et al. 2021, Zhang et al. 2024, Osorio-Ferero et al. 2021, Kjaerby et al. 2022). How ultraslow oscillations of different neuromodulators are related has been only recently explored (Zhang et al. 2024). In this study, dual recording of oxytocin (Oxt) and ACh with GRAB sensors showed that the levels of the two neuromodulators were indeed correlated at ultraslow frequencies with a 2 s temporal shift. Furthermore, this shift could be explained by a hippocampal-to-lateral septum intermediate pathway, in which the level of ACh causally impacts hippocampal activity, which then in turn controls Oxt levels. Given the known temporal relationship between ripples, ACh and Oxt, and now with our work, between ripples and 5-HT, one could infer the relative timing of ultraslow oscillations of ACh, Oxt and 5-HT. While dual recordings of norepinephrine (NE) and 5-HT have not been performed, a similar correlation with temporal shift could be hypothesized given the parallel relationships between NE and spindles (Osorio-Ferero et al. 2021), and 5-HT and ripples, with the known temporal delay between ripples and spindles (Staresina et al. 2023). The fact that the locus coeruleus receives particularly dense projections from the dorsal raphe nucleus (Kim et al. 2004) further suggests that 5-HT ultraslow oscillations could drive NE oscillations. How exactly ultraslow oscillations of serotonin are related to ultraslow oscillations of different neuromodulators in different brain regions remains to be studied.

We have further addressed this question and how it relates to the issue of causality in the Discussion section of the manuscript (p. 13):

“In addition to the difficulties involved with typical causal interventions already mentioned, the fact that the levels of different neuromodulators are interrelated and affected by ongoing brain activity makes it very hard to pinpoint ultraslow oscillations of one specific neuromodulator as controlling specific activity patterns, such as ripple timing. While a recent paper purported to show a causative effect of norepinephrine levels on ultraslow oscillations of sigma band power, the fact that optogenetic inhibition of locus coeruleus (LC) cells, but also excitation, only caused a minor reduction of the ultraslow sigma power oscillation suggests that other factors also contribute (Osorio-Ferero et al., 2021). Generally, it is thought that many neuromodulators together determine brain states in a combinatorial manner, and it is probable that the 5-HT oscillations we measure, like the similar oscillations in NE, are one factor among many.

Nevertheless, given the known effects of 5-HT on neurons, it is not unlikely that the 5-HT fluctuations we describe have some impact on the timing of ripples, MAs, hippocampal-cortical coherence, or EMG signals that correlate with either the rising or descending phase. In fact, causal effects of 5-HT on ripple incidence (Wang et al. 2015, ul Haq et al. 2016 and Shiozaki et al. 2023), MA frequency (Thomas et al. 2022), sensory gating (Lee et al. 2020), which is subserved by inter-areal coherence (Fisher et al. 2020), and movement (Takahashi et al. 2000, Alvarez et al. 2022, Jacobs et al. 1991 and Luchetti et al. 2020) have all been shown. Our added findings that serotonin affects ripple incidence in hippocampal slices in a dose-dependent manner (Figure S1) further suggests that the relationship between ultraslow 5-HT oscillations and ripples we report may indeed result, at least in part, from a direct effect of serotonin on the hippocampal network.

Whether these ‘causal’ relationships between 5-HT and the different activity measures we describe can be used to support a causal link between ultraslow 5-HT oscillations and the correlated activity we report remains an open question. To that point, some studies have described changes in ultraslow oscillations due to manipulation of serotonin signaling. Specifically, reduction of 5-HT_{1A} receptors in the dentate gyrus was recently shown to reduce

the power of ultraslow oscillations of calcium activity in the same region (Turi et al. 2024). Furthermore, psilocin, which largely acts on the 5-HT_{2a} receptor, decreased NREM episode length from around 100 s to around 60 s, and increased the frequency of brief awakenings (Thomas et al. 2022). While ultraslow oscillations were not explicitly measured in this study, the change in the rhythmic pattern of NREM sleep episodes and brief awakenings, or microarousals, suggests an effect of psilocin on ultraslow oscillations during NREM. Although these studies do not necessarily point to an exclusive role for 5-HT in controlling ultraslow oscillations of different brain activity patterns, they show that changes in 5-HT can contribute to changes in brain activity at ultraslow frequencies.”

A major question that has been left out from the study and discussion is how the same level of serotonin before and after the peak could be differentially related to the opposite observed phenomenon. What are the possible parallel mechanisms for distinguishing between the rising and falling phases? Any neurophysiological evidence for sensing the direction of change in serotonin concentration (or any other neuromodulator), and is there any physiological functionality for such mechanisms?

We have added a paragraph in the discussion to address how this differentiation of the 5-HT signal may be carried out (Discussion, paragraph #3, p. 10):

“In order for the ultraslow oscillation phase to segregate brain activity, as we have observed, the hippocampal network must somehow be able to sense the direction of change of serotonin levels. While single-cell mechanisms related to membrane potential dynamics are typically too fast to explain this calculation, a theoretical work has suggested that feedback circuits can enable such temporal differentiation, also on the slower timescales we observe (Tripp and Eliasmith, 2010). Beyond the direction of change in serotonin levels, temporal differentiation could also enable the hippocampal network to discern the steeper rising slope versus the flatter descending slope that we observe in the ultraslow 5-HT oscillations (Figure S2), which may also be functionally relevant (Cole and Voytek, 2017). The distinction between the rising and falling phase of ultraslow oscillations is furthermore clearly discernible at the level of unit responses, with many units showing preferences for either half of the ultraslow period (Figure S6). Another factor that could help distinguish the rising from the falling phase is the level of other neuromodulators, as it is likely the combination of many neuromodulators at any given time that defines a behavioral substate. Given the finding that ACh and Oxt exhibit ultraslow oscillations with a temporal shift (Zhang et al. 2024), one could posit that distinct combinations of different levels of neuromodulators could segregate the rising from the falling phase via differential effects of the combination of neuromodulators on the hippocampal network.”

Functionally, the ability to distinguish between the rising and falling phases of an oscillatory cycle is a form of phase coding. A well-known example of this can be seen in hippocampal place cells, which fire relative to the ongoing theta oscillations. The key advantage of phase coding is that it introduces an additional dimension, i.e. phase of firing, beyond the simple rate of neural firing. This allows for the multiplexing of information (Panzeri et al., 2010), enabling the brain to encode more complex patterns of activity. Moreover, phase coding is metabolically more efficient than traditional spike-rate coding (Fries et al., 2007).

Reviewer #2 (Public review):

Summary:

In their study, Cooper et al. investigated the spontaneous fluctuations in extracellular 5-HT release in the CA1 region of the hippocampus using GRAB5-HT3.0. Their findings revealed the presence of ultraslow frequency (less than 0.05 Hz) oscillations in 5-HT levels during both NREM sleep and wakefulness. The phase of these 5-HT oscillations was found

to be related to the timing of hippocampal ripples, microarousals, electromyogram (EMG) activity, and hippocampal-cortical coherence. In particular, ripples were observed to occur with greater frequency during the descending phase of 5-HT oscillations, and stronger ripples were noted to occur in proximity to the 5-HT peak during NREM. Microarousal and EMG peaks occurred with greater frequency during the ascending phase of 5-HT oscillations. Additionally, the strongest coherence between the hippocampus and cortex was observed during the ascending phase of 5-HT oscillations. These patterns were observed in both NREM sleep and the awake state, with a greater prevalence in NREM. The authors posit that 5-HT oscillations may temporally segregate internal processing (e.g., memory consolidation) and responsiveness to external stimuli in the brain.

Strengths:

The findings of this research are novel and intriguing. Slow brain oscillations lasting tens of seconds have been suggested to exist, but to my knowledge they have never been analyzed in such a clear way. Furthermore, although it is likely that ultra-slow neuromodulator oscillations exist, this is the first report of such oscillations, and the greatest strength of this study is that it has clarified this phenomenon both statistically and phenomenologically.

Weaknesses:

As with any paper, this one has some limitations. While there is no particular need to pursue them, I will describe ten of them below, including future directions:

(1) Contralateral recordings: 5-HT levels and electrophysiological recordings were obtained from opposite hemispheres due to technical limitations. Ipsilateral simultaneous recordings may show more direct relationships.

Although we argue that bilateral symmetry defines both the serotonin system and many hippocampal activity patterns (Methods: Dual fiber photometry and silicon probe recordings), we agree that ipsilateral recordings would be superior to describe the link between serotonin and electrophysiology in the hippocampus. In addition to noting that a recent study has adopted the same contralateral design (Zhang et al. 2024), we add a reference further supporting bilateral hippocampal synchrony, specifically of dentate spikes (Farrell et al. 2024). However, as functional lateralization has been recently proposed to underlie certain hippocampal functions in the rodent (Jordan 2020), future studies should ideally include both imaging and electrophysiology in a single hemisphere to guarantee local correlations rather than assuming inter-hemispheric synchrony. This could be accomplished using an integrated probe with attached optical fibers, as described in Markowitz et al. 2018, which is however technically more challenging and has, to our knowledge, not yet been implemented with fiber photometry recordings with GRAB sensors. Given the required separation of a few hundred micrometers between the probe shanks and the optical fiber cannula, it is important to consider whether the recordings are capturing the same neuronal populations. For example, there is a risk of recording electrical activity from dorsal hippocampal neurons while simultaneously measuring light signals from neurons in the intermediate hippocampus, which are functionally distinct populations (Fanselow and Dong 2009).

(2) Sample size: The number of mice used in the experiments is relatively small (n=6). Validation with a larger sample size would be desirable.

While larger sample sizes generally reduce the influence of random variability and minimize the impact of outliers on conclusions, our use of mixed-effects models mitigates these concerns by accounting for both inter-session and inter-mouse variability. With this

approach, we explicitly model random effects, such as the variability between individual mice and sessions, alongside fixed effects (such as treatment), which ensures that our results are not driven by random fluctuations in a few individual mice or sessions. Furthermore, the inclusion of random intercepts and slopes in the models allows for the possibility that different animals and/or sessions have different baseline characteristics and respond to different degrees of magnitude to the treatment. In summary, while validating these findings with a larger sample size would certainly help detect more subtle effects, we are confident in the robustness of the conclusions presented.

(3) Lack of causality: The observed associations show correlations, not direct causal relationships, between 5-HT oscillations and neural activity patterns.

We agree that the data we present in this study is largely correlational and generally avoid claims of causality in the manuscript. In the Discussion section, we discuss barriers to interpreting typical causal interventions in vivo, such as optogenetic activation of raphe nuclei: “The two previously mentioned in vivo studies showing reduced ripple incidence...” (paragraph #10, pg. 12), as well as an added section on further causality considerations in the Discussion section of the manuscript (paragraph #12, pg. 13): “In addition to the difficulties involved with...”

Due to these barriers, as a first step, we wanted to describe how physiological changes in serotonin levels are correlated to changes in the hippocampal activity. Equipped with a deeper understanding of physiological serotonin dynamics, future studies could explore interventions that modulate serotonin in keeping with the natural range of serotonin fluctuations for a given state. On that point, another challenge which we have not mentioned in the manuscript is that modulating serotonin, or any neuromodulator’s levels, has the potential, depending on the degree of modulation, to transition the brain to an entirely different behavioral state. This then complicates interpretation, as one is not sure whether effects observed are due to the changes in the neuromodulator itself, or secondary to changes in state. At the same time, 5-HT activity drives networks which in return can change the release of other neurotransmitters, leading to indirect effects.

The results of our in vitro experiments suggest that a causal relationship between serotonin and ripples is possible (Figure S1). Though the hippocampal slice preparation is clearly an artificial model, it provides a controlled environment to isolate the effects of serotonin manipulation on the hippocampal formation, without the confounding influence of systemic 5-HT fluctuations in other brain regions. Notably, the dose-dependent effects of serotonin (5-HT) wash-in on ripple incidence observed in vitro closely mirror the inverted-U dose-response curve seen in our in vivo experiments across states, where small increases in serotonin lead to the highest ripple incidence, and both lower and higher levels correspond to reduced ripple activity. This parallel suggests that the gradual washing of serotonin in our in vitro system may mimic the tonic firing changes in serotonergic neurons that occur during state transitions in vivo. These findings underscore the importance of studying how different dynamics of serotonin modulation can differentially affect hippocampal network activity.

(4) Limited behavioral states: The study focuses primarily on sleep and quiet wakefulness. Investigation of 5-HT oscillations during a wider range of behavioral states (e.g., exploratory behavior, learning tasks) may provide a more complete understanding.

We agree that future studies should investigate a broader range of behavioral states. For this study, as we were focused on general sleep and wake patterns, our recordings were done in the home cage, and we limited ourselves to the basic behavioral states described in the paper. Future studies should be designed to investigate ultraslow 5-HT oscillations during different behaviors, such as continuous treadmill running. Specifically, a finer segregation of extended

wake behaviors by level of arousal could greatly add to our understanding of the role of ultraslow serotonin oscillations.

(5) Generalizability to other brain regions: The study focuses on the CA1 region of the hippocampus. It's unclear whether similar 5-HT oscillation patterns exist in other brain regions.

Given the reported ultraslow oscillations of population activity in serotonergic neurons of the dorsal raphe nucleus (Kato et al. 2022) as well as the widespread projections of the serotonergic nuclei, we would expect a broad expression of ultraslow 5-HT oscillations throughout the brain. So far, ultraslow 5-HT oscillations have been described in the basal forebrain, as well as in the dentate gyrus, in addition to what we have shown in CA1 (Deng et al. 2024 and Turi et al. 2024). Furthermore, our results showing that hippocampal-cortical coherence changes according to the phase of hippocampal ultraslow 5-HT oscillations suggests that 5-HT can affect oscillatory activity either indirectly by modulating hippocampal cells projecting to the cortical network or directly by modulating the cortical postsynaptic targets. Given the heterogeneity in projection strength, as well as in pre- and postsynaptic serotonin receptor densities across brain regions (de Filippo & Schmitz, 2024), it would be interesting to see whether local ultraslow 5-HT oscillations are differentially modulated, e.g. in terms of oscillation power. Future studies investigating different brain regions via implantation of multiple optic fibers in different brain areas or using the mesoscopic imaging approach adopted in Deng et al. 2024, will be needed to examine the extent of spatial heterogeneity in this ultraslow oscillation.

(6) Long-term effects not assessed: Long-term effects of ultra-low 5-HT oscillations (e.g., on memory consolidation or learning) were not assessed.

While beyond the scope of our current study, we agree that an important next step would involve modulating the ultraslow serotonin oscillation after learning, and then examining potential effects on memory consolidation, presumably via changes in ripple dynamics, though many possibilities could explain potential effects. There, our results suggest it would be important to isolate effects due to the change in ultraslow oscillation features, rather than simply overall levels of 5-HT. To that end, it would be important to test different modulation dynamics, specifically modulating the oscillation strength, around a constant mean 5-HT level by carefully timed optogenetic stimulation/inhibition. Afterwards, showing a clear correlation between the strength of the 5-HT modulation and memory performance would be important to establishing the relationship, as done in Lecci et al 2017, where more prominent ultraslow oscillations of sigma power in the cortex during sleep, alongside a higher density of spindles, were correlated with better memory consolidation. Given the tight coupling of spindles and ripples during sleep, it is possible that a similar effect on memory consolidation would be observed following changes in ultraslow 5-HT oscillation power.

(7) Possible species differences: It's uncertain whether the findings in mice apply to other mammals, including humans.

We agree that the experiments should ultimately be replicated in humans. In the 2017 study by Lecci et al., the authors highlighted the shared functional requirements for sleep across species, despite apparent differences, such as variations in sleep volume. To explore these commonalities, the researchers conducted parallel experiments in both mice and humans, aiming to identify a universal organizing structure. They discovered that the ultraslow oscillation of sigma power serves this role, enabling both species to balance the competing demands of arousability and sleep imperviousness. Based on this finding, it is plausible that ultraslow oscillations of serotonin, which similarly modulate activity according to arousal levels, would serve a comparable function in humans.

(8) Technical limitations: The temporal resolution and sensitivity of the GRAB5-HT3.0 sensor may not capture faster 5-HT dynamics.

The kinetics of the GRAB5-HT3.0 sensor used in this study limit the range of serotonin dynamics we can observe. However, the ultraslow oscillations we measure reflect temporal changes on the scale of 20 s and greater, whereas the GRAB sensor we use has sub-second on kinetics and below 2 s off kinetics (Deng et al. 2024). Therefore, the sensor is capable of reporting much faster activity than the ultraslow oscillations we observe, indicating that the ultraslow 5-HT signal accurately reflects the dynamics on this time scale. Furthermore, the presence of ultraslow oscillations in spiking activity—observed in the hippocampal formation (Gonzalo Cogno et al., 2024; Aghajan et al., 2023; Penttonen et al., 1999) and in the dorsal raphe (Mlinar et al., 2016), which are not affected by the same temporal smoothing, suggests that the oscillations we record are not likely due to signal aliasing, but instead reflect genuine oscillatory activity. Of course, this does not preclude that other, faster serotonin dynamics are also present in our signal, some of which may be too fast to be observed. For instance, rapid serotonin signaling via the ionotropic 5-HT_{3a} receptors could be missed in our recordings. Additionally, with the fiber photometry approach we adopted, we are limited to capturing spatially broad trends in serotonin levels, potentially overlooking more localized dynamics.

(9) Interactions with other neuromodulators: The study does not explore interactions with other neuromodulators (e.g., norepinephrine, acetylcholine) or their potential ultraslow oscillations.

We agree that the interaction between neuromodulators in the context of ultraslow oscillations is an important issue, which we have addressed in our response to reviewer #1 under ‘Weaknesses.’

(10) Limited exploration of functional significance: While the study suggests a potential role for 5-HT oscillations in memory consolidation and arousal, direct tests of these functional implications are not included.

We agree and reference our answer to (6) regarding memory consolidation. Regarding arousal, direct tests of arousability to different sensory stimuli during different phases of the ultraslow 5-HT oscillation during sleep would be beneficial, in addition to the indirect measures of arousal we examine in the current study, e.g. degree of movement (icEMG) and long range coherence. In line with what we have shown, Cazettes et al. (2021) has demonstrated a direct relationship between 5-HT levels and pupil size, an indicator of arousal level, which like our findings, is consistent across behavioral states.

Reviewer #3 (Public review):

Summary:

The activity of serotonin (5-HT) releasing neurons as well as 5-HT levels in brain structures targeted by serotonergic axons are known to fluctuate substantially across the animal's sleep/wake cycle, with high 5-HT levels during wakefulness (WAKE), intermediate levels during non-REM sleep (NREM) and very low levels during REM sleep. Recent studies have shown that during NREM, the activity of 5HT neurons in raphe nuclei oscillates at very low frequencies (0.01 - 0.05 Hz) and this ultraslow oscillation is negatively coupled to broadband EEG power. However, how exactly this 5-HT oscillation affects neural activity in downstream structures is unclear.

The present study addresses this gap by replicating the observation of the ultraslow oscillation in the 5-HT system, and further observing that hippocampal sharp wave-

ripples (SWRs), biomarkers of offline memory processing, occur preferentially in barrages on the falling phase of the 5-HT oscillation during both wakefulness and NREM sleep. In contrast, the raising phase of the 5-HT oscillation is associated with microarousals during NREM and increased muscular activity during WAKE. Finally, the raising 5-HT phase was also found to be associated with increased synchrony between the hippocampus and neocortex. Overall, the study constitutes a valuable contribution to the field by reporting a close association between raising 5-HT and arousal, as well as between falling 5-HT and offline memory processes.

Strengths:

The study makes compelling use of the state-of-the-art methodology to address its aims: the genetically encoded 5-HT sensor used in the study is ideal for capturing the ultraslow 5-HT dynamics and the novel detection method for SWRs outperforms current state-of-the-art algorithms and will be useful to many scientists in the field. Explicit validation of both of these methods is a particular strength of this study.

The analytical methods used in the article are appropriate and are convincingly applied, the use of a general linear mixed model for statistical analysis is a particularly welcome choice as it guards against pseudoreplication while preserving statistical power.

Overall, the manuscript makes a strong case for distinct sub-states across WAKE and NREM, associated with different phases of the 5-HT oscillation.

Weaknesses:

All of the evidence presented in the study is correlational. While the study mostly avoids claims of causality, it would still benefit from establishing whether the 5-HT oscillation has a direct role in the modulation of SWR rate via e.g. optogenetic activation/inactivation of 5-HT axons. As it stands, the possibility that 5-HT levels and SWRs are modulated by the same upstream mechanism cannot be excluded.

We agree that causality claims cannot be made with our data, and acknowledge the interest in exploring causal interactions between ultraslow serotonin oscillations and the correlated activity we measure. We address this point in depth in our answer to Reviewer #2, Weaknesses #3.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

One major question in the presented data is the nature of the asymmetrical shape of the targeted slow events. How much does it reflect the 5-HT concentration and how much is this shape affected by the dynamics of the designed 5-HT sensor? This needs to be addressed in more detail referencing the original paper for the used sensor.

We have added a paragraph in the Results section of the manuscript to address the asymmetric waveform of the ultraslow 5-HT oscillations and whether it could be affected by the asymmetric kinetics of the GRAB sensor we use: “The waveform of these ultraslow 5-HT oscillations...” (Results, paragraph #4, pg. 5). We include an extended answer to the question here:

Indeed, the GRAB5-HT3.0 sensor we use in the study shows activation response kinetics which are faster than their deactivation time, with time constants at 0.25 s and 1.39 s, respectively (Deng et al. 2024). Likewise, the slope of the rising phase of the ultraslow serotonin oscillation we measure is faster than the slope of the falling phase, and the ratio of time spent in the rising phase versus the falling phase is less than 1, indicating longer falling

phases (Figure S2). Although we cannot completely rule out that the asymmetric shape of the ultraslow serotonin oscillations we record is affected by this asymmetry in the 5-HT sensor kinetics, we believe this is unlikely, as the 5-HT signal clearly contains reductions in 5-HT levels that are much faster than the descending phase of the ultraslow oscillation. Although it is difficult to directly compare the different-sized signals, the reported timescales of off kinetics, on the order of a few seconds (Deng et al. 2024), are far below the tens of seconds timescale of the ultraslow oscillation. Furthermore, the finding that some dorsal raphe neurons modulate their firing rate at ultraslow frequencies, and moreover that all examples of such ultraslow oscillations shown display clear asymmetry in rising time versus decay, suggests that the asymmetry we observe in our data could be due to neural activity rather than temporal smoothing by the sensor (Mlinar et al. 2016). In this same direction, another study found similar asymmetry in extracellular 5-HT levels measured with fast scan cyclic voltammetry (FSCV), a technique with greater temporal resolution (sampling rate of 10 Hz) than GRAB sensors, after single pulse stimulation (Bunin and Wightman 1998). In this study, 5-HT was shown to be released extrasynaptically, making the longer clearing time compared to the release time intuitive. Finally, the observation that the onsets and offsets of ripple clusters, recorded with a sampling rate of 20 kHz, are precisely aligned with the peaks and troughs of ultraslow serotonin oscillations (Figure 1, H1-2, columns 2-3) suggests that the duration of the falling phase is not artificially distorted by the temporal smoothing of the sensor dynamics.

Regardless of the dynamics of the serotonin concentration, it should be noted that the elicited neuronal effect might have different dynamics compared to the 5-HT concentration that need to be more studied: to address this one can either examine the average of the broadband LFP (not high passfiltered by the amplifier) or the distribution of simultaneously recorded spiking activity around the peak of ultra-slow oscillations.

We have added Figure S6, showing unit activity relative to the phase of ultraslow serotonin oscillations.

From this analysis, we uncover three groups of units which are largely preserved across states (Figure S6, E vs. F), albeit with a slight temporal shift rightward from NREM to WAKE (Figure S6, C vs. D). Namely, some units spike preferentially during the rising phase, some during the falling phase, and a third group have no clear phase preference. Unit activity during the falling phase is unsurprising, as it is where ripples largely occur, which themselves are associated with spike bursts. During the rising phase, the unit activity we observe could correspond to firing of the hippocampal subpopulation known to be active during NREM interruption states (Jarosiewicz et al. 2002, Miyawaki et al. 2017). While the units' phase preference was tested based on the category of rising vs. falling phase, as this division described most variation in the data, a few units in the 'No preference' group showed heightened activity near the oscillation peak. However, given the very small number of units with this preference, more unit data is needed to describe this group, ideally with high-density recordings. Overall, most units showed a falling vs. rising phase preference, indicating a phase coding of hippocampal activity by 5-HT ultraslow oscillations.

Related to the previous point, it would be helpful to show the average cycle shape of these oscillations (relative to the phase 0 extracted in Figure 3) and do the shape comparison across sessions and also wake/NREM

We agree, and to this end we have added Figure S2. From this waveform analysis, we show that the ultraslow serotonin oscillation is asymmetric, with the rising phase having a greater slope, but shorter length, than the falling phase. While this asymmetry is observed both in NREM and WAKE, the slope difference and length ratio difference in rising vs. falling phase is greater in NREM (Figure S2. B).

In Figure 3D, there seem to be oscillatory rhythms with faster cycles on top of the targeted oscillations. That would make the phase estimation less accurate, e.g. in the left panel, in the second cycle, it is not clear if there are two faster cycles or it is one slow cycle as targeted, and if noted in the rising phase of the second fast cycle there are no ripples. This might suggest that regardless of specific oscillation frequency whenever 5-HT is started to get released, the ripples are suppressed and once the 5-HT is not synaptically effective anymore the ripples start to get generated while the photometry signal starts to wane with the serotonin being cleared. Still, if there is any rhythmicity between bouts of no ripple, it would suggest an ultra-slow regularity in the 5-HT release.

The reviewer is correct to point out that some faster increases in serotonin, which occur on top of the ultraslow oscillations we measure, seem to be associated with decreased ripple incidence, as in the example referenced. The dominance of ultraslow frequencies in the power spectrum of the 5-HT signal suggests, however, that oscillations faster than the ultraslow oscillations we describe are far less prevalent in the data. While there may be some coupling of ripples and other measures to serotonin oscillations of different frequencies, this may be hard or impossible to detect with phase analysis based on their infrequent occurrence and nonstationary nature. In fact, we show in Figure S3 that the strongest phase modulation of ripples by ultraslow serotonin oscillations is observed in the frequencies we use (0.01-0.06 Hz). Methodologically, phase analysis indeed assumes stationary signals, which are rare if not absent in physiological data (Lo et al. 2009), however generally the narrower the frequency band, the better the phase estimation. The narrow frequency band we use provides phase estimates that are largely robust and unaffected by the presence of faster oscillations, as can be seen in the example phase traces shown in Figure 4.

The hypothesis that the rising phase burst of synaptic serotonin is what silences ripples, and that with the clearing of serotonin from the synapses, ripples recover, is a possible explanation of our findings. However, if this were the case, one could expect the ripple rate to increase over the course of the falling phase of ultraslow 5-HT oscillations, as 5-HT decreases, and peak at the trough. This is at odds with what we observe, namely a fairly uniform distribution of ripples along the falling phase (Figure 3F2,F4). Furthermore, the Mlinar et al. 2016 study describes a subpopulation of raphe neurons whose firing rates themselves oscillate at ultraslow frequencies, rather than on-off bursting at ultraslow frequencies, which would argue against this hypothesis. However, as this study looks at a small number of neurons in slices, further in vivo experiments examining firing rates of median raphe neurons are required to understand how the ultraslow oscillation of extracellular serotonin that we measure is generated as well as how it is related to ripple rates.

In Figure 3B, it is not clear why IRI is z-scored. It would be informative to have the actual value of IRI. What is the z relative to? Is it the mean value of IRI in each recording session? Is this to reduce the variability across sessions?

We have now included in Figure 3D a box plot displaying the IRI distributions across different states and sessions. To minimize inter-session variability, data were z-scored within each session for visualization purposes. However, all general linear models were based on raw data, and as a result, the raw differences in IRI are shown in Figure 3C.

Figure 3E, panel labels don't match with the caption

We are grateful to the reviewer for pointing out this mistake, which we have corrected in the updated version of the manuscript.

In the text related to Figure 3E, the related analysis can be more clearly described. "phase preference of individual ripples" does not immediately suggest that the occurring phase of each ripple relative to the targeted oscillation is extracted. I suggest performing this analysis individually for each session and summarizing the results across the sessions.

We have reworded the sentence in Results: 5-HT and ripples to better reflect the analysis performed: "Next, we calculated the ultraslow 5-HT phases at which individual ripples occurred during both NREM and WAKE (3E-F) ...". Regarding session-level data, we have added Figure S3, which shows session level mean phase vectors, as well as the grand mean across sessions for both NREM and WAKE. Included in this figure are session level means for frequency bands outside of the ultraslow band we used in our study, intended to show that ripples are most strongly timed by the ultraslow band (0.01-0.06 Hz), reflected by the greater amplitude of the mean phase vector for this band.

Figure 3E2, based on the result of ripple-triggered 5-HT in left panels of 2H1-2, one would expect to see a preferred phase closer to 180 (toward the end of the falling phase), it would be helpful to compare and discuss the results of these two analyses.

The reviewer is correct to point out the apparent discrepancy in where the mean ripple falls with respect to the ongoing serotonin oscillation between the two figures mentioned. We have addressed this point in Results: 5-HT and ripples, paragraph #4: "This result appear to be at odds with...".

Regarding the analysis in 3F, please also compare the power distribution of ripples between NREM and wake. This will help to better understand the potential difference behind the observed difference: how much the strong ripples are comparable between wake and NREM. It is also necessary to report the ripple detection failure rate across ripples with different strengths.

We have added a figure showing analysis done on a subset of the data in which ripples were manually curated in order to evaluate the performance of the ripple detection model (Figure S7) and explanatory text in Methods: Model performance: "To ensure that our model ...". In summary, while missed ripples did tend to have lower power than correctly detected ripples, including them did not change the distribution of ripples by the phase of the ultraslow serotonin oscillation (Figure S7C). We would also note that while the phase preference is noisier than what is presented in Figure 3F because this analysis was done with a small subset of all recorded ripples, the fact that ripples occur more clearly on the falling phase is visible for both detected ripples and detected + false negative ripples.

The mixed-effects model examining the influence of 5-HT ultraslow oscillation phase on ripple power revealed no significant effect of state ($p = 0.088$). This indicates that whether the data were collected during NREM or wake periods did not significantly impact ripple power and that the lack of a significant effect (in Figure 3G,H) in WAKE is probably not due to a difference in the distribution of ripple power between states.

4D, y label is z?

We are grateful for the reviewer to point that out, yes, the y label should be 'z-score', as the two traces represent z-scored 5-HT (blue) and z-scored shuffled data (orange). Figure 4D2 and Figure 2H1-2, which show similar data, have been corrected to address this oversight.

Relating to Figure 4, EMG comparison across phases of the oscillations is insightful. Two related and complementary analyses are to compare the theta and gamma power between the falling and rising phases.

We have addressed this suggestion in Figure S5 A-C. While low gamma, high gamma and theta power are modulated identically in NREM, with higher power observed during the falling phase than the rising phase, during WAKE, different patterns can be seen. Specifically, low gamma power shows no phase preference, while high gamma shows a peak near the center of the ultraslow 5-HT oscillation. Theta power, as in NREM, is higher during the falling phase of ultraslow 5-HT oscillations. Increased power across many frequency bands was shown to coincide with decreases in DRN population activity during NREM, which matches with what we report here (Kato et al. 2022). In summary, while NREM patterns are consistent in all frequency bands tested, aligning with the pattern of ripple incidence, in WAKE low and high gamma power show different relationships to ultraslow 5-HT phase.

In the manuscript, we have used the data in both Figure S5 and S6 (unit activity relative to ultraslow 5-HT oscillations), to argue against the idea that our coherence findings result from a lack of activity in the rising phase (see next question), which would have the effect of ‘artificially’ reducing coherence in the falling phase relative the rising phase. The text can be found in Results: 5-HT and hippocampal cortical coherence, paragraph #2.

The results presented in Figure 5 could be puzzling and need to be further discussed: if the ripple band activity is weak during the rising phase, in what circumstances the coherence between cortex and CA1 is specifically very strong in this band?

As mentioned in the previous answer, we have addressed this concern in Results: 5-HT and hippocampal-cortical coherence, paragraph #2. In summary, it is true that the higher coherence in rising phase than in the falling phase for the highest frequency band (termed ‘high frequency oscillation’ (HFO), 100-150 Hz) could be unexpected, given that ripples occur largely during the falling phase. A few points could help explain this finding. Firstly, it should be noted that power in the 100-150 Hz band can arise from physiological activity outside of ripples, such as filtered non-rhythmic spike bursts (Liu et al. 2022), whose coherent occurrence in the rising phase could explain the coherence findings. Secondly, coherence is a compound measure which is affected by both phase consistency and amplitude covariation (Srinath and Ray 2014), thus from only amplitude one cannot predict coherence. Furthermore, HFO power in the cortex is highest near the peak of ultraslow 5-HT oscillations (Figure S5D), as opposed to the falling phase peak in the hippocampus. This shows a lack of covariation in amplitude by phase between the hippocampus and cortex at this frequency band. An alternative explanation of our findings regarding coherence could be that in the rising phase, there is simply little to no activity, which is easier to ‘synchronize’ than bouts of high activity. Hippocampal unit activity in the rising phase (Figure S6) suggests however, that it is not likely to be the absence of activity supporting higher coherence in the rising phase across frequencies. Additional experiments using high density recordings should be conducted to examine 5-HT ultraslow oscillations and their role in gating activity across brain regions, though these results strongly suggest some role exists.

Reviewer #2 (Recommendations for the authors):

I would like to offer two comments. I believe that these are not unusual requests, and thus I would like the authors to respond.

(1) It would be prudent to investigate the possibility that the observed correlation between ultraslow and hippocampal ripples/microarousals is merely superficial and that there are unidentified confounding factors at play. For example, it would be beneficial to

provide evidence that administering a serotonin receptor inhibitor result in the disappearance of the slow oscillation of ripples and microarousals, or that the correlation with ultraslow is no longer present. Please note that the former experiments do not require GRAB5-HT3.0 imaging.

We agree that causality claims cannot be made with our data and acknowledge the interest in exploring causal interactions between ultraslow serotonin oscillations and the correlated activity we measure. We address this point in depth in our answer to Reviewer #2, Weaknesses #3. We would further like to note that given the large number of serotonin receptors and the lack of selectivity of many serotonin receptor antagonists, a pharmacological approach would be difficult, though the results certainly useful. Finally, we highlight the psilocin study, which reported changes in the rhythmic occurrence of microarousals, and therefore likely ultraslow oscillations, after administering a 5-HT_{2a} receptor agonist, suggesting a potential causal effect of 5-HT (via 5-HT_{2a} receptor) on MA occurrence (Thomas et al. 2022).

(2) The slow frequency appears to be associated with the default mode network as observed in fMRI signals. The neural basis of the default mode network remains unclear; therefore, a more detailed examination of this possibility would be beneficial.

We agree that it would be interesting to investigate the role of 5-HT in the neural basis of the DMN.

The DMN as described in humans (Raichle et al. 2001) and rodents (Lu et al. 2012) may indeed include some parts of the hippocampus and perhaps some of our neocortical recordings could also be considered part of the DMN. The fact that the activity across the inter-connected brain structures of the DMN is correlated at ultraslow time scales (Gutierrez-Barragan et al. 2019, Mantini et al. 2007), as well as serotonin's ability to modulate the DMN is intriguing (Helmbold et al. 2016). Further studies simultaneously recording DMN activity via fMRI and electrical activity via silicon probes, as done in Logothetis et al. 2001, could elucidate further a potential link between ultraslow oscillations and the DMN, with serotonergic modulation as a means to understand any potential contribution of serotonin.

Reviewer #3 (Recommendations for the authors):

(1) The impact of the study would benefit from an experiment causally testing the effect of hippocampal 5-HT levels on hippocampal physiology, e.g. using optogenetic manipulations.

We agree that causality claims cannot be made with our data and acknowledge the interest in exploring causal interactions between ultraslow serotonin oscillations and the correlated activity we measure. We address this point in depth in our answer to Reviewer #2, Weaknesses #3.

(2) Data presentation: the figures are of poor resolution, making some diagram details and, more importantly, some example traces (e.g. Figure 1A, right) impossible to see. This should be corrected by either increasing figure resolution or making important figure elements large enough to be readable.

We apologize for the poor resolution and have corrected it in the updated version of the manuscript.

(3) Differences in some figure panels are not statistically assessed: Figure 1H (differences in spectrum peak power), Figure 3E1 & Figure 3E3 (directional bias of the circular distributions), Figure 4C (difference from 0 mean).

We acknowledge this oversight and have added statistical tests for all three figures, as well as further information regarding the models used in Methods: Statistics.

(4) Lines 279-280: the claim that the study shows "organization of activity by ultraslow oscillations of 5-HT" implies a causal role of 5-HT in organizing hippocampal activity. I suggest that this statement be toned down to reflect the correlational nature of the presented evidence.

We have rephrased the sentence in question to the following: "In our study, including both NREM and WAKE periods allowed us to additionally show that the temporal organization of activity relative to ultraslow 5-HT oscillations operates according to the same principles in both states...", which we believe better reflects the temporal correlation we describe.

(5) While the study claims to use the EMG (i.e. electromyograph) signal, it does not describe any electrodes placed inside the muscle in the methods section. The SleepScoreMaster toolbox used in the study estimates the EMG using high-frequency activity correlated across recording channels, so I assume this is how this signal was obtained. While such activity may well reflect muscular noise to some degree, it is an indirect measure as the electrodes are not in the muscle. Since the EMG signal is central to the message of the manuscript, the method for calculating it should be described in the methods section and it should be explicitly labelled as an indirect measure in the main text, e.g. by referring to this signal as pseudo-EMG.

We agree and have added explanatory text to the State Scoring subsection in Methods. Given that the EMG we refer to is derived from intracranial data, and not from traditional EMG probes, we now refer to the EMG as intracranial EMG, or icEMG for short, throughout the main text.

(6) Is ripple frequency or ripple duration different across the rising and falling phases of the ultraslow oscillation?

We have now investigated this suggestion in Figure S4, where we show that ripple frequency is higher in the falling phase than rising phase, while ripple duration appears to show no phase preference.

(7) Lines 315-317: I am not sure why the manuscript refers to the coupling between EMG and 5-HT levels as 'puzzling' given that, as stated, the locomotion-inducing effects of 5-HT are well documented. While the fact that even non-locomotory motor activity may be associated with 5-HT rise is certainly interesting (although not sure if 'puzzling'), the manuscript does not directly compare the association of 5-HT levels with locomotory and non-locomotory EMG spikes. Thus, I think this discussion point is not fully warranted.

We agree and have rephrased the discussion point in question to reflect that the EMG link to serotonin oscillations is not necessarily surprising, given both the literature linking 5-HT and spontaneous movement in the hippocampus, as well as the involvement of 5-HT in repetitive movements, where the role for a regularly-occurring oscillation is perhaps more intuitive.

(8) Line 441: Reference #67 does not describe the use of fiber photometry.

The reviewer is to correct to point out this typo, which has been now corrected. The reference in question should be 64, where fiber photometry experiments are described. For further clarity, we have changed our referencing scheme to include authors and years in in-text references.

(9) In Figures 3E1-3, the phase has different bounds than in the other Figures in the manuscript (0:360 vs -180:180), this should be corrected for consistency.

We agree and have made changes so that all figures have a phase range of -180 to 180°.

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<https://doi.org/10.7554/eLife.101105.2.sa0>